Conduct: Market-level and Firm-level (Homogeneous Goods)

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Introduction

- Conduct refers to how firms behave (what model of competition they are playing).
- In these notes we will discuss:
  * Identification.
  * Testing/Estimation.
  * Applications.

Introduction

- 2 types of conduct:
  * Horizontal: How firms behave with competitors? (Cournot, Bertrand, Monopoly, Stackelberg)
  * Vertical: How firms behave with other upstream/downstream (double marginalization, nonlinear pricing, RPM)
- In particular, horizontal conduct can exist downstream (selling) or upstream (buying).
- Most of the time, when we say conduct, we will refer to horizontal conduct.

Roadmap: Market-Level Papers

- These notes move from **market-level** to **firm-level** data, and from **assuming** conduct to **estimating** it.

Paper	Data	MC	Main question	Ex.
Bresnahan (1982)	Market	Unknown	How to identify λ?	1
Genesove–Mullin (1998)	Market	Known	How robust is $\hat{\lambda}$ to functional form?	—
Clay–Troesken (2003)	Market	Known	Same as above for whiskey	—
Corts (1999)	Market	—	Why $\hat{\lambda}$ can mislead even with good data	3
Bergquist et al. (2020)	Market	—	RCT passthrough to infer conduct without FOC	5

Running thread: can we separately identify conduct and MC?

Roadmap: Firm-Level Papers

- Moving to firm-level data adds cost heterogeneity, richer instruments, and the ability to study mergers and cartel stability.

Paper	**Data**	**MC**	**Main question**	**Ex.**
Firm-level framework	Firm	Both	General Ω estimation and merger analysis	2
Igami (2015)	Firm	Recovered	Cartel impact when collusion dates are known	4
Bergquist et al. (2020)	Firm	—	RCT passthrough to infer conduct without FOC	5
Igami (2022)	Firm	Recovered	Incentives to collude and cartel stability	6

Running thread: firm-level data does not solve the core identification problem — it provides better instruments and richer applications.

Historical Background

- Horizontal conduct is adequate to answer a long-lasting question: How to identify *collusive* behavior?
- Horizontal conduct allows different models to be described by a **conduct parameter**.
- Estimate a conduct parameter and find out what type of game is being played.
- There is not theoretical support for a conduct parameter except in a few specific scenarios.
- Therefore, **conduct parameters do not have a structural interpretation**. In other words, are atheoretical.[1]
- We will refer to conduct parameters as *internalization parameters* or measuring deviations from competition.

[1] For more detail see [Reiss and Wolak, 2007] (Sections 5 and 6).

Main Difficulty

- We don't observe reliable values of marginal costs.
- We also don't know if firms collude or compete.
- We are asking our methods to estimate BOTH.
- Therefore, things are easier if:
  * We know periods of competition/collusion and/or
  * We have good measures of marginal costs.

Identification: Market-level [Bresnahan, 1982]

- Given **market** level data: (Q, P, W, Y).
- Demand Function

$$Q = D(P, Y, \alpha) + \epsilon \tag{1}$$

- Assume a continuous of models between perfect competition and collusion. Then assume a generalized first-order condition

$$P = MC(Q, W, \beta) - \lambda h(Q, Y, \alpha) + \eta \tag{2}$$

  with $MR = P + \lambda h(Q, Y, \alpha)$.

- Note that perfect competition ($\lambda = 0$) or collusion $\lambda = 1$.[2]
- We could estimate (1)-(2) simultaneously or (1) and later (2). But are things this simple?

[2] Note that under Cournot $\lambda = 1/N$

Identification: Market-level [Bresnahan, 1982]

- We know how to estimate a demand. In this context, we could use as instrument a cost shifter (W).

- Assume demand and MC are linear. Then,

$$\begin{aligned} Q =& \alpha_0 + \alpha_1 P + \alpha_2 Y + \epsilon \\ MC =& \beta_0 + \beta_1 Q + \beta_2 W \end{aligned}$$

  which results in the equivalent equation (2)

$$P = \beta_0 + \gamma Q + \beta_2 W + \eta \tag{3}$$

  with $\gamma = \beta_1 - \lambda/\alpha_1$.

Identification: Market-level [Bresnahan, 1982]

- Similarly to demand estimation, there is an endogeneity issue in equation (3) related to P and Q.
- We need an instrument for Q. Let's use Y (demand shifter).
- Is this enough to identify λ? NO.
- γ depends on $\alpha_1, \beta_1, \lambda$. α_1 is not a problem, but we cannot separately identify β_1 and λ.

Identification: Market-level [Bresnahan, 1982]

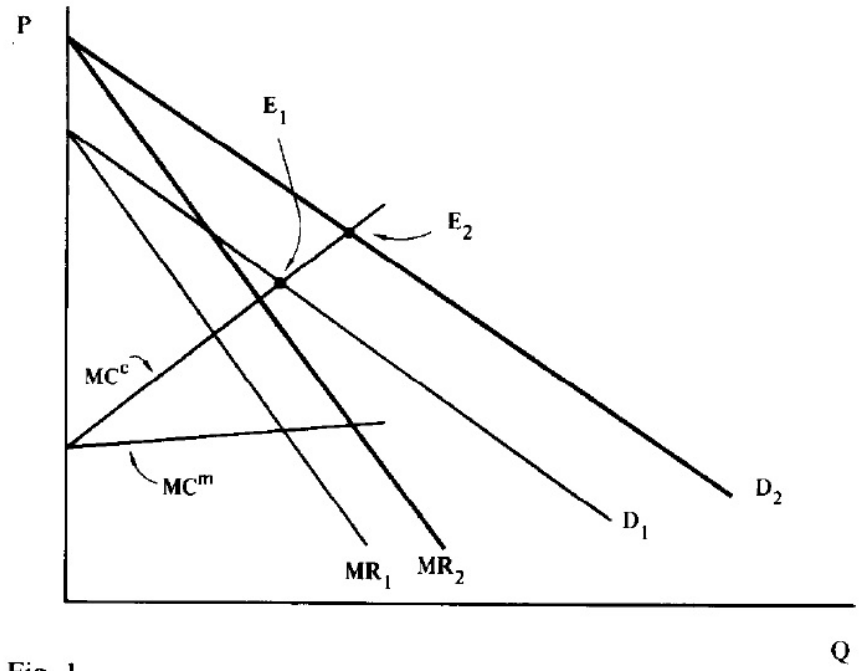


Fig. 1.

We cannot differentiate between equilibrium under competition ($P = MC^c$) or monopoly ($MR = MC^m$)!

Identification: Market-level [Bresnahan, 1982]

- Bresnahan's solution: Demand Rotators.[3]
  - Change demand equation to

$$Q = \alpha_0 + \alpha_1 P + \alpha_2 Y + \alpha_3 PZ + \alpha_4 Z + \epsilon$$

with Z as a new demand-side exogenous variable.

- Now, change (3) to

$$\begin{aligned} P &= \left(-\frac{\lambda}{\alpha_1 + \alpha_3 Z}\right) Q + \beta_0 + \beta_1 Q + \beta_2 W + \eta \\ P &= \lambda Q^* + \beta_0 + \beta_1 Q + \beta_2 W + \eta \end{aligned} \tag{4}$$

- We can use Y and Z to identify Q^* and Q.

[3] More general conditions in [Lau, 1982]

Identification: Market-level [Bresnahan, 1982]

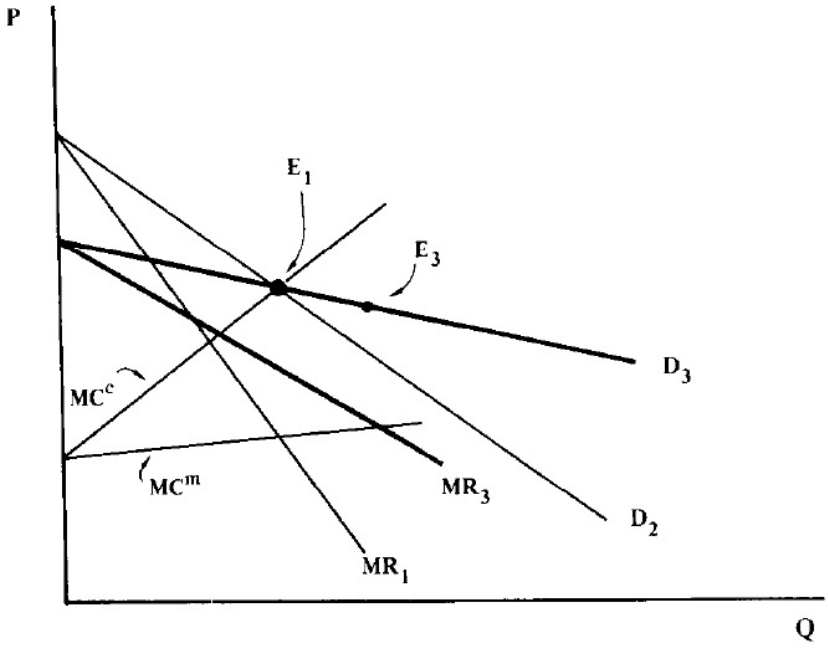


Fig. 2.

Rotations have no effect under competition but they do under a different model.

[Bresnahan, 1982]: Numerical Example — Setup

- **Demand** (level shifter Y, rotator Z):

$$Q = 12 - P + 5Y + 0.5PZ - Z \quad \Rightarrow \quad P = A(Y, Z) - c(Z) \cdot Q$$

Y	Z	A	c
0	0	12	1
1	0	17	1
0	1	22	2
1	1	32	2

- **Marginal cost:** $MC = \beta_0 + \beta_1 Q = 2 + Q$.

[Bresnahan, 1982]: Numerical Example — Setup

- **Equilibrium condition** $MR = MC$, where $MR = A - c(1+\lambda)Q$:

$$A - c(1+\lambda)Q = 2 + Q$$

Rearranging, this gives the **equilibrium pricing condition**:

$$P = \beta_0 + \underbrace{(\beta_1 + \lambda c)}_{\gamma} Q \qquad \Rightarrow \qquad \boxed{\gamma \equiv \beta_1 + \lambda c}$$

γ is the slope of P on Q in equilibrium. It combines the MC slope (β_1) and the conduct markup (λc). The researcher can estimate γ from data, but cannot separately recover λ and β_1 without additional variation in c.

- When $\lambda = 0$: $\gamma = \beta_1$, $P = MC$ (competition). When $\lambda = 1$: $\gamma = \beta_1 + c$ (monopoly).

- **True parameters** (unknown to researcher): $\lambda = 0.5$, $\beta_1 = 1$.

[Bresnahan, 1982]: $Z = 0$ — Level Shift Does Not Identify λ

- With $Z = 0$, $c = 1$. Equilibrium condition becomes:

$$A - (1 + \lambda)Q = 2 + Q \;\Rightarrow\; A - 2 = (2 + \lambda)Q$$

- Two equilibria from varying Y:

Y	A	Q^*	P^*	MR	MC
0	12	4	8	6	6
1	17	6	11	8	8

- The researcher recovers the slope of the equilibrium pricing condition from these two observations:

$$\hat{\gamma}_1 = \frac{\Delta P}{\Delta Q} = \frac{11 - 8}{6 - 4} = 1.5 \qquad \text{where } \gamma = \beta_1 + \lambda c$$

- **Not identified:** $\hat{\gamma}_1 = \beta_1 + \lambda \cdot 1 = 1.5$ has infinitely many solutions. Any (λ, β_1) pair summing to 1.5 is consistent with the data — $(0,\ 1.5)$, $(0.5,\ 1)$, $(1,\ 0.5)$, and everything in between.

[Bresnahan, 1982]: $Z = 1$ — Rotator Identifies λ

- With $Z = 1$, $c = 2$. Equilibrium condition becomes:

$$A - 2(1 + \lambda)Q = 2 + Q \;\Rightarrow\; A - 2 = (3 + 2\lambda)Q$$

- Two equilibria from varying Y:

Y	A	Q^*	P^*	MR	MC
0	22	5	12	7	7
1	32	7.5	17	9.5	9.5

[Bresnahan, 1982]: $Z = 1$ — Rotator Identifies λ

- The researcher recovers the slope under $Z = 1$:

$$\hat{\gamma}_2 = \frac{17 - 12}{7.5 - 5} = 2.0$$

- **Now λ is identified.** The rotation changed c from 1 to 2 and $\hat{\gamma}$ changed from 1.5 to 2.0. Since $\gamma = \beta_1 + \lambda c$ and β_1 does *not* depend on c, the entire change in $\hat{\gamma}$ came from λ reacting to Δc:

$$\Delta\hat{\gamma} = \hat{\gamma}_2 - \hat{\gamma}_1 = 0.5 \;=\; \lambda \cdot \Delta c = \lambda \cdot 1 \;\Rightarrow\; \hat{\lambda} = 0.5 \checkmark$$

$$\hat{\beta}_1 = \hat{\gamma}_1 - \hat{\lambda} \cdot c_1 = 1.5 - 0.5 = 1.0 \checkmark$$

$\Rightarrow$ **The rotator works because it moves γ through λ only. A level shift moves neither.**

Identification: Market-level [Bresnahan, 1982]

- Note that conduct identification **will depend on functional form of demand and marginal costs**.
- Alternatively we could have impose constant marginal costs ($\beta_1 = 0$).

Insights from [Reiss and Wolak, 2007]

- Note that λ implies[4] (if we had all these elements!)

$$\lambda = (-\eta^D)\frac{(P - MC)}{P} \tag{5}$$

  with η^D as the price-elasticity of demand.

- Argues that conduct identification is plague with mispecification on marginal costs plus untested demand functional forms.

- This highlights that λ will depend on MC and η^D.

- Additionally, note that if we impose a specific λ (and given a η^D estimate) we can recover MC and potentially estimate it.

[4] Also called the adjusted Lerner-Index

Insights from [Reiss and Wolak, 2007]

- 2 extra points:
  1. **All this is at the market level!** Assumes λ is constant for all firms.
     1.1 Does λ depends on demand or supply shocks? What theory says? [Green and Porter, 1984, Rotemberg and Saloner, 1986]
  2. λ identified by functional form (our demand really needs a rotator?)
     2.1 Are we forcing misspecification?

[Genesove and Mullin, 1998]

- Tests conduct parameter approach using U.S. East Coast cane sugar refining industry (at the end of XIX century).
- How robust are our estimates of λ to different demand functional form (assuming we know MC)?
  * Demand estimation uses Cuban raw sugar imports as IV (cost-shifter).
  * Demand estimation under different functional forms.
  * MC is known (they can estimate it under $c = c_0 + kP_{raw}$).
- With these elements then (5) can be used to find λ.

[Genesove and Mullin, 1998]

TABLE 4 **Demand for Refined Sugar, Derived Estimates**

	(1) Quadratic	(2) Linear	(3) Log-Linear	(4) Exponential	(5) Lerner Index
$P^M(c)$: *High Season* = 0	$2.57 + .67c$	$2.91 + .5c$	$1.97c$	$1.89 + c$	
$P^M(c)$: *High Season* = 1	$3.96 + .67c$	$3.96 + .5c$	$10.1c$	$3.85 + c$	
$P(c;\theta)$: *High Season* = 0	$\frac{7.72\theta}{2+\theta} + \frac{2}{2+\theta}c$	$\frac{5.82\theta}{1+\theta} + \frac{1}{1+\theta}c$	$\frac{2.03}{2.03-\theta}c$	$1.89\theta + c$	
$P(c;\theta)$: *High Season* = 1	$\frac{11.88\theta}{2+\theta} + \frac{2}{2+\theta}c$	$\frac{7.91\theta}{1+\theta} + \frac{1}{1+\theta}c$	$\frac{1.10}{1.10-\theta}c$	$3.85\theta + c$	
η at Full sample mean					
High Season = 0	2.18	2.24	2.03	2.13	
High Season = 1	1.03	1.04	1.10	1.05	
Adjusted Lerner index, L_η					**(Unadjusted)**
Mean	.099	.107	.095	.097	.054
Standard Deviation	.097	.118	.083	.089	.045
Standard Error	.024	.028	.021	.022	.0046

$\lambda \approx 0.10$ (Adjusted Lerner Index). **Potentially low because domestic producers afraid of imports.** Larger during high demand periods.

[Genesove and Mullin, 1998]

- Now, let's assume we don't know MC.
  * Fix a demand estimation functional form (linear).
  * Apply NLIV or GMM to estimate obtain λ and β (parameters of MC): $\mathbb{E}\left[\left(MR(\alpha, \lambda) - MC(\beta)\right) Z\right]$.
  * Z includes a constant, high demand dummy and Cuban imports (why?).

[Genesove and Mullin, 1998]

TABLE 7 **NLIV Estimates of Pricing Rule Parameters**

	Linear		Direct Measure
	(1)	(2)	(3)
$\hat{\theta}$	.038 (.024)	.037 (.024)	.10
$\hat{c}_o$	.466 (.285)	.39 (.061)	.26
$\hat{k}$	1.052 (.085)		1.075

λ is underestimated when we don't know MC. This is something stressed by the Corts' Critique (later).

[Genesove and Mullin, 1998]

- Finally, what if we assume a specific λ and try to recover MC?
  * Fix a demand estimation functional form (linear).
  * Fix a level of λ and estimate MC.

[Genesove and Mullin, 1998]

TABLE 9 Cost and Price Estimates for Different Behavioral Models

	Perfect Competition		Cournot I		Cournot II		Monopoly		Direct Measure
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
$\hat{c}_o$	.674	.476	.00	.069	.00	.00	.00	.00	.26
	(.281)	(.034)	(.239)	(.071)	(.922)	(.400)	(1.65)	(.563)	
$\hat{k}$	1.015		1.096		.883		.529		1.075
	(.087)		(.071)		(.253)		(.471)		
Predicted price changes, Cuban Revolution									
$\widehat{\Delta P}$	.689	.729	.620	.608	.300	.365	.179	.365	.702
	(.059)		(.040)		(.086)		(.086)		

Notes: Demand parameters are taken from the linear form in Table 4, estimated separately by season. Cost parameters are constrained to be nonnegative. Predicted increase in refined prices is based upon the increase in the price of raw sugar by 68 cents per hundred pounds from the third quarter of 1896 to the third quarter of 1897.

The models that are more competitive are better at recovering MC.

[Genesove and Mullin, 1998]

TABLE 10 **NLIV Estimates of Pricing Rule Parameters, Price War Regimes**

	Linear		Direct Measure
	(1)	(2)	(3)
$\hat{\theta}_0$	.046 (.023)	.046 (.023)	.129
$\hat{\theta}_1 - \hat{\theta}_0$	−.112 (.069)	−.108 (.051)	−.127
$\hat{c}_o$	.375 (.280)	.402 (.056)	.26
$\hat{k}$	1.084 (.083)		1.075

Better fit when λ is allowed to change over time (and we don't know MC).

[Genesove and Mullin, 1998]

- λ can be estimated and although no the same, it is in the neighborhood of observed values.
- Demand functional form not that important **(if we know MC)**.
- Additional MC information does not improve λ estimates.
- Flexible functional form (and/or) time-varying λ improves recovering MC.

[Clay and Troesken, 2003]

- Does exactly the same as [Genesove and Mullin, 1998] but for whiskey (also at the end of XIX century).
- Differently than sugar, there are less observations in whiskey so they rely on data imputation (for Q).
- This paper is another test on the conduct parameter approach.
- As in sugar, it is an easy-to-obtain marginal costs industry.

[Clay and Troesken, 2003]

TABLE V
PRICING RULES, ELASTICITY, AND THE LERNER INDEX

	Quadratic	Linear	Loglinear	Exponential	Lerner (unadjusted)
$P(c,\theta)$ 16 obs.	$(2.89\theta+2c)/(2+\theta)$	$(2.13\theta+c)/(1+\theta)$	$1.74c/(1.74\text{-}\theta)$	$0.77\theta+c$	
$P(c,\theta)$ 38 obs.	$(2.69\theta+2c)/(2+\theta)$	$(2.02\theta+c)/(1+\theta)$	$1.99c/(1.99\text{-}\theta)$	$0.67\theta+c$	
η at full mean 16 obs.	− 1.557	− 1.476	− 1.738	− 1.634	
η at full mean 38 obs.	− 1.762	− 1.670	− 1.992	− 1.874	
Elasticity-adjusted Lerner 16 obs.	0.081 (0.049)	0.076 (0.047)	0.090 (0.055)	0.084 (0.051)	0.052 (0.033)
Elasticity-adjusted Lerner 38 obs.	0.091 (0.057)	0.086 (0.054)	0.103 (0.065)	0.097 (0.061)	0.052 (0.033)

Notes: Standard errors are in parentheses.

Demand estimation functional form (again) does not seem to be important for price-elasticity **when MC is known**.

[Clay and Troesken, 2003]

TABLE VII
NONLINEAR ESTIMATES OF PRICING RULE PARAMETERS

	Linear 38 obs	Linear 38 obs	Quadratic 38 obs	Quadratic 38 obs	Log-Linear 38 obs	Log-Linear 38 obs	Direct
θ	0.173**	0.120**	0.346***	0.241***	0.346***	0.241***	0.090
	(0.046)	(0.036)	(0.092)	(0.073)	(0.092)	(0.073)	
c_0	−0.058**	−0.029	−0.176***	−0.110**	0.291***	0.215***	0.00
	(0.029)	(0.025)	(0.053)	(0.050)	(0.081)	(0.049)	
k	0.216**		0.216***		0.216***		0.23
	(0.0490)		(0.0490)		(0.049)		

Notes: Standard errors are in parentheses. Significance levels are indicted with *'s, where *** indicates significance at the 1 percent level, ** indicates significance at the 5 percent level, and * indicates significance at the 10 percent level.

But results are sensitive to it when don't know MC (Table VII).
About the latter, λ is overestimated.

[Clay and Troesken, 2003]

TABLE VIII
NONLINEAR ESTIMATES OF PRICING RULE PARAMETERS, TWO PRICING REGIMES

38 obs	Linear I	Linear II	Direct
θ_0	0.085**	0.067*	0.078
	(0.035)	(0.038)	
θ_1	0.176***	0.153***	0.176
	(0.036)	(0.039)	
c_0	− 0.024	0.0012	0.00
	(0.032)	(0.0290)	
k	0.252***		0.23
	(0.037)		

Notes: Standard errors are in parentheses. Significance levels are indicted with *'s, where *** indicates significance at the 1 percent level, ** indicates significance at the 5 percent level, and * indicates significance at the 10 percent level. Prices were considered high in the following quarters: 1883:3, 1885:3, 1886:3, 1888:3-4, 1892:4-1893:1.

A flexible approach to λ makes values almost the same as the observed ones!

Uses (market level)

- The main point of these models was to measure (at the market level) competition/collusion.
- In those years (80s-90s) disaggregated data was not readily available. **Nowadays we have WAY MORE data.**
- Research at the market level is not a frontier topic but help us understand the benefits/shortcomings of the conduct approach.
- Note that λ only makes sense in certain specific values!!!
- Something not stressed by the previous papers is that we can get counterfactual market outcomes under different λ values.

Exercise 1: Motivation

- We estimated $\hat{\lambda}$. Now ask: *what would market outcomes look like under a different conduct?*
- **Key:** demand is held fixed. Replacing $\hat{\lambda}$ with λ^{CF} changes only the markup firms add on top of MC — the demand curve does not move.
- **Logic:** lower $\lambda^{CF} \Rightarrow$ lower markup $\Rightarrow$ lower P^{CF} and higher Q^{CF}.

$\Rightarrow$ **What you learn:** how much of the observed price is conduct (markup) versus MC — the welfare cost of market power.

Exercise 1: Different conduct values (market level)

1. Replace $\hat{\lambda}$ with λ^{CF} in (3). Also replace P with $\hat{P} = \hat{F}(Q)$ (estimated demand function).

$$\hat{P}(Q^{CF}) = \hat{\beta}_0 + \left(\hat{\beta}_1 - \lambda^{CF}/\hat{\alpha}_1\right) Q^{CF} + \hat{\beta}_2 W$$

2. Solve for Q^{CF} (root-finding in 1-d).
3. Therefore we can obtain different market outcomes depending on λ^{CF}.
4. Let's define market outcomes (prices, quantities, etc) as $O(\lambda^{CF})$.

[Corts, 1999]

- This is a great paper! Also known as *Corts Critique.*

- Under BH1989 approach

$$
\begin{aligned}
P =& c_i'(q_i) - \theta_i P'(Q) q_i \\
\tilde{\theta}_i =& \frac{P - c_i'}{-P' q_i} = \frac{P - c_i'}{P} N\epsilon
\end{aligned}
$$

  with $\tilde{\theta}$ measuring equilibrium **values/levels.**

- Setup:

$$
\begin{aligned}
P(Q_t; x_t) =& a_0 + a_1 x_t + a_2 Q_t + \epsilon_t \\
c_i'(q_i) =& c_0 + c_1 w_t
\end{aligned}
$$

  which results in $P_t = \beta_0 + \beta_1 w_t + \beta_2 q_{it} + \epsilon_{it}$ with $\beta_2 = -\theta_i a_2$

[Corts, 1999]

- Asymptotic estimates

$$\hat{\beta}_2 = \frac{a_1}{\gamma} + Na_2$$

$$\hat{\theta}_i = \frac{\hat{\beta}_2}{-\hat{a}_2} = -\frac{a_1}{a_2\gamma} - N$$

- Estimated conduct $\hat{\theta}$ determined by demand parameters and γ (reaction of eq. quantity to demand shifter x). Based on equilibrium **variation.**

- OTOH, $\tilde{\theta}$ measures equilibrium **values/levels** (does not depend on **variation** in x)

- If we assume that optimal rule $q_i^*(.)$ is linear in x, then $\gamma = dq_i^* / dx_t$.

[Corts, 1999]

- In other words, Equation (5) measures equilibrium **values/levels**, but estimation of λ based on equilibrium **variation.**
- Only when both are the same (Proposition 1) there are no problems in the conduct estimation.
- When this happens? Let's define a stylized repeated game to check what exactly the conduct estimation is doing.
- Uses a simulated repeated game under duopoly (symmetric Cournot) subject to demand shocks.

[Corts, 1999]

- Inverse demand has already been estimated: $P_t = A_t - b_t Q_t$.
- Supply (to estimate γ, λ), allow **cost shock** ε_t^S:

$$P_t \;=\; x_t^c \gamma \;+\; \lambda_t\, b_t Q_t \;+\; \varepsilon_t^S, \qquad w_t := b_t Q_t.$$

- Endogeneity: w_t co-moves with ε_t^S (supply shifts move P_t, Q_t).
- Instruments Z_t must satisfy:

  Relevance: $\mathrm{Cov}(Z_t, w_t) \neq 0$, **Exogeneity:** $\mathrm{Cov}(Z_t, \varepsilon_t) = 0$.

- *Demand instruments* (e.g., shifters of A_t) satisfy these under standard exogeneity; *cost instruments* can also be valid and useful.

[Corts, 1999]

Partial out observed costs via Frisch–Waugh:

$$v_t := M_x P_t, \quad u_t := M_x w_t, \quad \tilde{\varepsilon}_t^S := M_x \varepsilon_t^S, \quad (\ M_x \text{ projects off } x_t^c\ ).$$

Then

$$v_t \ = \ \lambda\, u_t \ + \ \tilde{\varepsilon}_t^S.$$

With instruments Z_t, 2SLS estimates

$$\widehat{\lambda}^{IV} \ \to \ \lambda^\star = \frac{\mathrm{Cov}\big(\Pi_Z[v_t],\, \Pi_Z[u_t]\big)}{\mathrm{Var}\big(\Pi_Z[u_t]\big)} = \underbrace{\frac{\mathrm{Cov}\big(\Pi_Z[\lambda_t u_t],\, \Pi_Z[u_t]\big)}{\mathrm{Var}\big(\Pi_Z[u_t]\big)}}_{\text{“levels vs. slopes” term}} + \underbrace{\frac{\mathrm{Cov}\big(\Pi_Z[\tilde{\varepsilon}_t^S],\, \Pi_Z[u_t]\big)}{\mathrm{Var}\big(\Pi_Z[u_t]\big)}}_{\text{exclusion error}}.$$

Here $\Pi_Z[\cdot]$ denotes the projection on the instrument space.

[Corts, 1999]

- **Constant conduct:** $\lambda_t \equiv \lambda_0$.
- **Valid instruments:** $Z_t \perp \varepsilon_t^S \Rightarrow$ exclusion error term $= 0$.

$$\Rightarrow \quad \lambda^\star \;=\; \lambda_0.$$

Interpretation: only in this special case does the estimated slope coincide with the theoretical *level* implied by the FOC.

[Corts, 1999]

Even if $Z_t \perp \varepsilon_t^S$ (demand instruments purge the cost shock), the first term generally differs from λ_0:

1. **State-varying conduct** λ_t:

$$\lambda^\star = \frac{\mathrm{Cov}\big(\Pi_Z[\lambda_t u_t],\, \Pi_Z[u_t]\big)}{\mathrm{Var}\big(\Pi_Z[u_t]\big)} = \mathbb{E}[\lambda_t \omega_t]\,,\ \omega_t \propto \frac{\Pi_Z[u_t]^2}{\sum_s \Pi_Z[u_s]^2}.$$

   High-demand states (large $\Pi_Z[u_t]$) receive more weight $\Rightarrow$ *state-weighted slope*, not a level.

[Corts, 1999]

Even if $Z_t \perp \varepsilon_t^S$ (demand instruments purge the cost shock), the first term generally differs from λ_0:

2. **Elasticity variation tied to instruments**: if b_t rotates with the shifter, $u_t = M_x(b_t Q_t)$ blends level & slope of demand; using Q_t instead of $b_t Q_t$ aggravates this.
3. **Cost misspecification beyond** ε_t^S: capacity/overtime/convexity not spanned by x_t^c leak into $\tilde{\varepsilon}_t^S$; if $\tilde{\varepsilon}_t^S$ co-moves with $\Pi_Z[u_t]$, exclusion error $\neq 0$.
4. **Dynamics/expectations**: continuation values or inventory shadow costs vary with demand states; the static FOC is misspecified, and $\lambda^\star$ absorbs dynamic markups (pseudo-conduct).
5. **Generated-regressor noise in** b_t: Step-1 error correlated with Z_t tilts $\lambda^\star$.

[Corts, 1999]

- Let $\pi(q; x_t, w_t) = q\,(P(Nq; x_t) - c_0 - c_1 w_t)$ be profit when q is chosen at state x_t.
- Best-response profits $\pi^b(q; x_t, w_t) = \max_{q_0} q_0\,(P((N-1)q + q_0) - c_0 - c_1 w_t)$.
- Punishments profits are given by reversion to Nash (in period k) equal to $\pi^c(x_k, w_k)$.

[Corts, 1999]

- Then, an efficient equilibrium quantity is defined by

$$
\begin{aligned}
q^*(x_t, w_t, \delta) = & \text{argmax}_q \ \pi(q; x, w) \\
& \text{subject to } \pi b(q; x_t, w_t) \\
& + \sum_{i=1}^{\infty} \delta^i E_t \pi^c(x_{t+i}, w_{t+1}) \leq \pi(q; x_t, w_t) \\
& + \sum_{i=1}^{\infty} \delta^i E_t \pi(q^*(x_{t+i}, w_{t+i}, \delta); x_{t+i}, w_{t+i})
\end{aligned}
$$

- With a maximand (for an unconstrained max; IC does not bind)

$$
q^m(x_t, w_t) = \frac{a_0 + a_1 x_t - c_0 - c_1 w_t}{-2Na_2}
$$

[Corts, 1999]

- If it does (constrained max; IC does bind)

$$\tilde{q}(x_t, w_t, \delta) = \frac{a_0 + a_1 x_t - c_0 - c_1 w_t}{-(N+1)a_2} - \frac{2\sqrt{-a_2 L(x_t)}}{-(N+1)a_2}$$

with

$$L(x_t) = \sum_{i=1}^{\infty} \delta^i E_t[\pi(q^*(x_{t+i}, w_{t+i}, \delta); x_{t+i}, w_{t+i}) - \pi^c(x_{t+i}, w_{t+1})]$$

- Simulation results over different δ and demand shock persistence.

- Main message:
  * Only in specific cases the conduct estimation is reliable.
  * Conduct estimation **could underestimate actual conduct.** and sensitive to demand shock persistence.

[Corts, 1999]

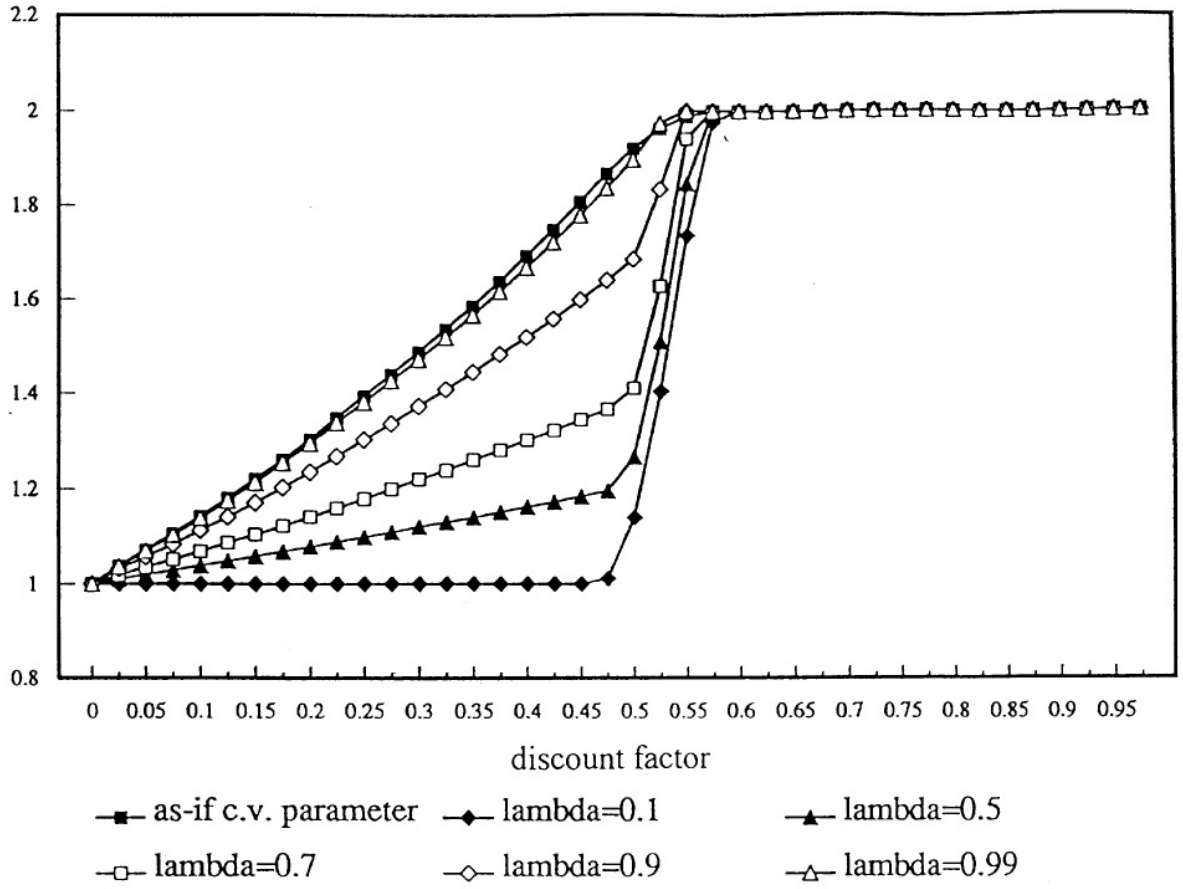
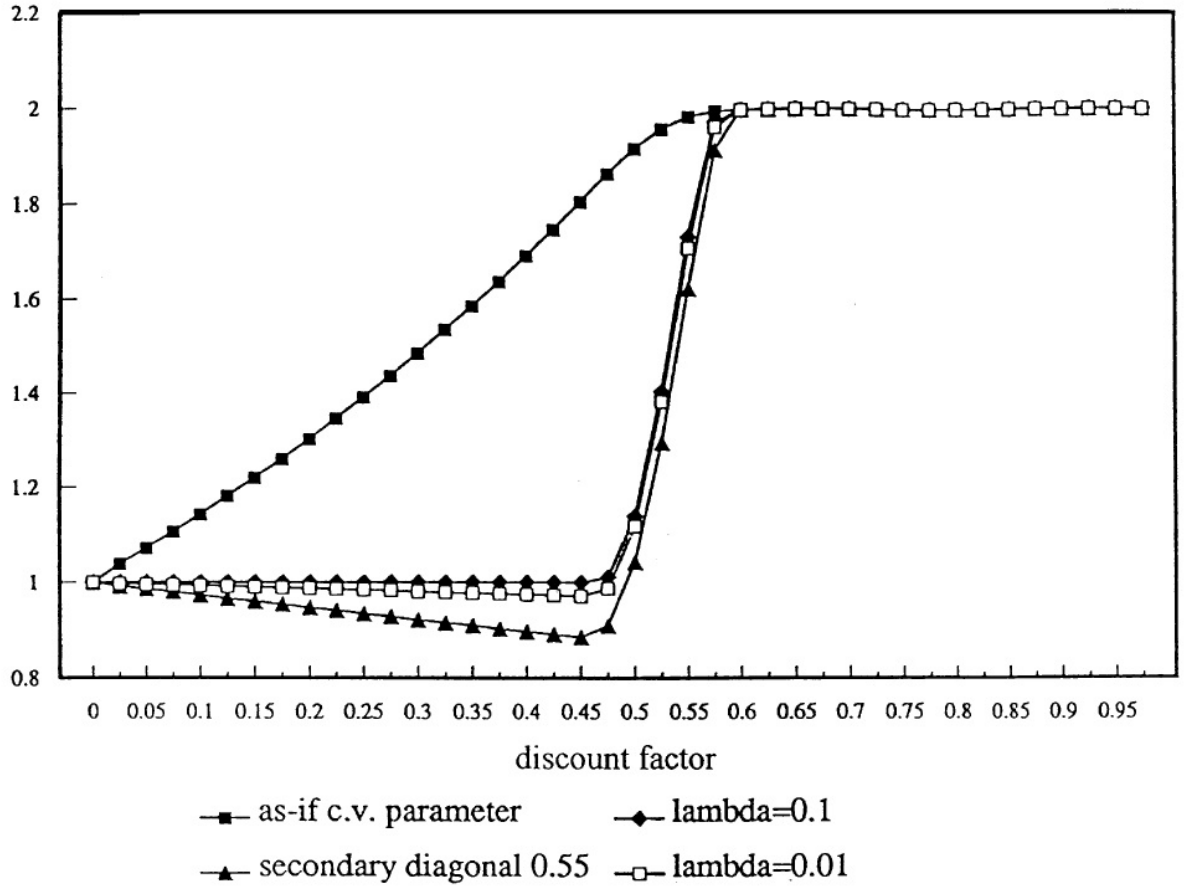


Fig. 2. Conduct parameters with positive serial correlation in demand.

Under monopolization they match (independently of the demand process; Proposition 2). With positive correlation, only when these are persistent.

[Corts, 1999]



When demand shocks have a negative correlation, conduct estimation misses actual values.[5]

[5](Warning) Check first paragraph in pg. 240

[Corts, 1999]

- Extremely influential paper and huge warning for conduct estimation.
- Note that this is a paper about biases **(underestimation)** and theoretical consistency.
- **If possible**, inclusive information about marginal costs.
- [Corts, 1994] (chapter 1) shows that if we impose specific structure in the demand shock, it is possible to **link** the theoretical and the estimated conduct parameter.
  - This idea was discarded in the published version ([Corts, 1999]).
  - But this link is something worthwhile to explore.

Exercise 2: Motivation

- Corts (1999) shows analytically that $\hat{\theta}$ can diverge from $\tilde{\theta}$ — but how large is the bias in practice?

- **Simulation answers this** in a controlled setting where the true $\tilde{\theta}$ is known, so we can compute $\hat{\theta} - \tilde{\theta}$ directly.

- **Two things to vary:**
  * δ higher $\Rightarrow$ collusion more stable $\Rightarrow$ less state-dependence $\Rightarrow$ smaller bias.

  * Demand shock persistence: iid shocks are the worst case; verify this numerically.

$\Rightarrow$ **What you learn:** the bias is quantitatively important for moderate δ and negatively correlated shocks — empirically plausible in many industries.

Exercise 2: Simulating the Corts Critique

- Simulate x_t and w_t as i.i.d. with K states.
- Set a specific value of δ.
- Set values of the previous states and demand and cost parameters such that all $q^* > 0$.
- Define a guess of $L^i(x_t)$
  1. Using $L^i(x_t)$ get $\tilde{q}(x_t, w_{t,\delta})$.
  2. Using $\tilde{q}(x_t, w_{t,\delta})$ get $L^{i+1}(x_t)$.
  3. Iterate until convergence of $L^i(x_t) - L^{i+1}(x_t)$.
- Using simulated values to make conduct estimation.

Identification: Firm-level (General Setting)

- Assume N firms playing a Cournot game with market power in the product good.[6]
- **We are setting the problem and estimation at the firm-level data. But for this, we need variation at that level!**
  - Difficult to find cost-shifters at the firm-month level.
  - Options: Institutional variation at firm level? Rival MC? Exports! This is why you need to know your industry!!!
- If we don't have firm-level variation still doable but estimation loses power.

[6] It can also be done with buyer market power.

Identification: Firm-level (General Setting)

- N firms
- Demand: $P = f(Q, Y, Z)$
- Profit:

$$\pi_i = (P - MC_i)q_i + \sum_{k \in N} \lambda_{ik} \pi_{-i}$$

  such that λ_{ik} is *how much firm i cares about firm k.*

- FONC:

$$P + P' q_i + \sum_{k \neq i \in N} \lambda_{ik} P' q_{-i} = MC_i$$

  and note that $\lambda_{ii} = 1$.

Identification: Firm-level (General Setting)

- Note that if $\lambda_{ik} = 0$ we return to a Cournot model.
- Additionally, λ_{ik} is called an internalization parameter but has no structural interpretation (besides $\lambda_{ik} = 0$ or $\lambda_{ik} = 1$).
- It is extremely difficult to estimate λ_{ik} for all and each (i, k) pair. Therefore, some assumptions will need to be made.
- Again, what variation could be used to estimate λ_{ik}?
  * We need firm-level variation (o.w. just estimate at market-level).
  * Only sources of variation are location of firms, rival marginal costs or other industry specific variables.
  * Not written here but λ_{ik} could also vary with time.

Identification: Firm-level (General Setting)

- For a specific month (market), we can then define an N-system of equations.

- Define Ω as an N-dimensional matrix. If firms are competing the it will be diagonal. If there is some conduct, then there are λ_{ik} in the off-diagonal.

$$\Omega = \begin{bmatrix} 1 & \lambda_{12} & \lambda_{13} & \dots & \lambda_{1N} \\ \lambda_{21} & 1 & \lambda_{23} & \dots & \lambda_{2N} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \lambda_{N1} & \lambda_{N2} & \lambda_{N3} & \dots & 1 \end{bmatrix}$$

- In vector form

$$\boldsymbol{P} + (\Omega \cdot P')\boldsymbol{q} = \boldsymbol{MC}$$

with $\boldsymbol{P}$, $\boldsymbol{q}$ and $\boldsymbol{MC}$ as N-1 vectors.

Exercise 3: Motivation

- We have $\hat{\lambda}_{ik}$ and $\widehat{MC}_i$. A merger between d and e changes ownership structure and potentially conduct for all firms.

- **Why not compare pre- and post-merger prices?** Demand, costs, and entry all change simultaneously — a naive comparison conflates conduct with everything else.

- **Counterfactual instead:** hold demand and costs fixed, remove the merger, solve for equilibrium. The difference is the causal effect of the merger.

$\Rightarrow$ **What you learn:** whether price increases come from fewer competitors (unilateral) or softer rivalry (coordinated), and whether efficiency gains offset either.

Exercise 2: Merger Policy

- Once we estimate at firm-level, we can discuss the effect of merger policy.

- We are not going to review all the merger literature here but we can summarize the effects of a merger in
  * Unilateral effects.
  * Coordinated effects.
  * Cost efficiencies.

- A merger review is an ex-ante permission when *the merger involves large firms.* A competition authority needs to measure the previous effects and approve or deny a merger.

- Our firm-level estimation allow us to say something? Let's assume we want to understand the effects of a merger between (d, e). Say our sample includes a pre and a post merger period.

Exercise 3: Merger Policy

- To incorporate all this into our framework we would need to consider the estimation of extra parameters.
- To make things simple, let's assume that all non-diagonal values in Ω are the same as λ.
- Then, the merger generates the following changes
  * Unilateral effects: $\lambda_{de}^{pre} = \lambda^{pre}$ to $\lambda_{de}^{post} = 1$.
  * Coordinate effects: λ^{pre} to λ^{post}.
  * Cost efficiencies: MC_d^{pre} and MC_e^{pre} to MC_d^{post} and MC_e^{post}.

Exercise 3: Merger Policy

- To measure the effect of the merger, let's undo the merger.

$$\hat{\boldsymbol{P}}(\boldsymbol{q}^{CF}) + (\Omega(\hat{\lambda}^{pre}_{de}, \hat{\lambda}^{pre}) \cdot \hat{P}'(\boldsymbol{q}^{CF}))\boldsymbol{q}^{CF} = \boldsymbol{MC}^{pre}$$

  and we solve for a N-1 $\boldsymbol{q}^{CF}$ vector (root-finding).

- Notice that this is a post-estimation exercise and only to be done using post-merger data.

Exercise 3: Merger Policy

- Unilateral effects refers to (ceteris paribus) the impact of lower competition. Intuitively, N to $N-1$ firms. This should increase market power.

- Coordinated effects refers to (ceteris paribus) how the merger changes the internalization of rivals profits. This can increase or decrease market power.

- Can we disentangle these effects? YES.
  * Unilateral effects: $O(\hat{\lambda}_{de}^{post}, \hat{\lambda}^{pre}, \boldsymbol{MC}^{post})$ - $O(\hat{\lambda}_{de}^{pre}, \hat{\lambda}^{pre}, \boldsymbol{MC}^{pre})$.
  * Coordinate effects: $O(\hat{\lambda}_{de}^{post}, \hat{\lambda}^{post}, \boldsymbol{MC}^{post})$ - $O(\hat{\lambda}_{de}^{post}, \hat{\lambda}^{pre}, \boldsymbol{MC}^{post})$.

Exercise 3: Merger Policy

- Notice that we have done an analysis ex-post.
- But what would you do if you would need to do a merger analysis ex-ante?

[Igami, 2015]

- Study of an international cartel in the coffee market between $1960 - 1989$.
- Consumer countries allowed producer countries to collude in order to make indirect transfers during the Cold War.
- What was the impact of this collusion episode?
- Notice that the exact dates of collusion/competition are known.
  * In certain years, firms compete a la Cournot ($\lambda_{ik} = 0$).
  * Although collusive periods are known, it is not known if collusion was the maximal or not ($\lambda_{ik} \in (0, 1]$)?
- Additionally, in this market there is an exogenous player (Vietnam) whose production is not entirely driven by economic profit.

[Igami, 2015]

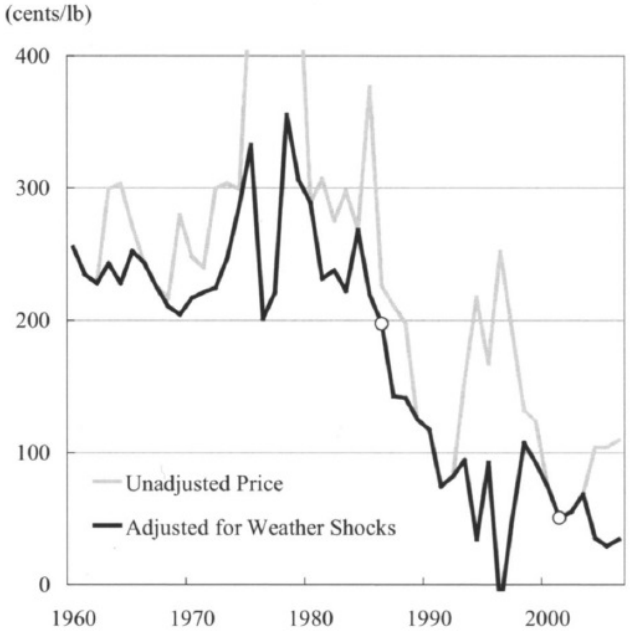


Figure 1
Real Price of Coffee Beans

Notes: The prices are in the 2007 constant U.S. dollar. The Adjusted price plots the residual from regressing the original Unadjusted price on the weather shock variable, which is explained in section 3.

[Igami, 2015]

- World Demand $Q_t = f(P_t, X_t, Z_t, \epsilon_t)$.
- Vietnam is a fringe exporter (does not collude).
- Timing
  1. World demand coffee beans with supply $Q_t = q_t^V + Q_t^C$.
  2. Vietnam exports entire harvest q_t^V.
  3. Given $P(Q_t)$ and q_t^V, cartel members (54) decide their exports collectively Q_t^c.

[Igami, 2015]

- Cournot model with N countries. Country i in month t chooses export q_{it} to maximize profits.

$$\pi_{it} = P(Q_t^C, q_t^V)q_{it} - C(q_{it}, W_{it})$$

- Cost shifter W_{it} affects marginal costs mc_{it}.

- FOC is given by: $P_t + P' q_{it} = mc_{it}$

[Igami, 2015]

TABLE II
ESTIMATES OF WORLD COFFEE BEAN DEMAND

	OLS			IV		
Dep. var.: P_t (Coffee Price)	[1]	[2]	[3]	[4]	[5]	[6]
Q_t (Coffee Exports)	−75.6***	−200.6***	−194.4***	−284.1***	−329.7***	−292.7***
	(14.2)	(57.4)	(32.7)	(79.7)	(73.0)	(36.9)
X_t (Buyers' GDP)		357.2**	616.6***		679.9***	854.7***
		(141.4)	(111.7)		(183.6)	(85.6)
Z_t (Tea Price)			.5484***			.5329***
			(.1225)			(.0908)
Constant	656***	1271***	922***	1797***	1915***	−1422***
	(90)	(296)	(164)	(425)	(369)	(212)
R^2	.36	.50	.67	–	.37	.59
Adjusted Number of observations	576	576	576	576	576	576

Notes: The IV estimation uses weather shocks as the instruments for Coffee Exports. Column [4] does not report R-squared, because it is negative. ***, **, and *indicate significance at the 0.1%, 1%, and 5% levels, respectively. Standard errors are in parentheses (clustered by calendar year).

Demand estimation with IVs like weather and/or producer prices and returns a price-elasticity of -0.15 using the full sample (and at market level).

[Igami, 2015]

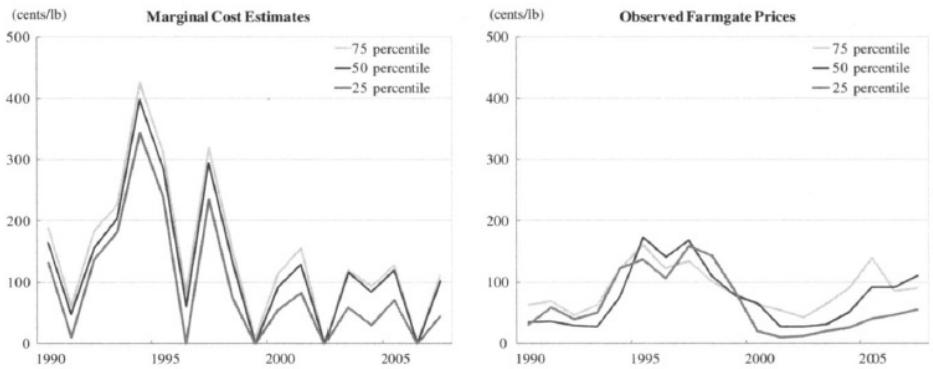


Figure 2
Marginal Cost Estimates and Data
Notes: The figure compares the estimated (left) and observed (right) measures of marginal costs, by plotting those of the countries at specific quartiles based on the 1990–2006 average.

At country-level we impose a Cournot model FOC only during the post-collusion period to obtain an estimate of MC_i.

[Igami, 2015]

TABLE III
WELFARE ANALYSIS OF THE INTERNATIONAL COFFEE AGREEMENT

Annualized 1979–88 average	[A] Actual (with Cartel)	[B] Counterfactual (without Cartel)	[C] Difference ([A]–[B])
(1) Price (cents/lb)	313.8	179.6 [169.8, 189.4]	134.2 [144.0, 124.4]
(2) World Exports (million bags)	64.0	69.6 [70.0, 69.2]	−5.5 [−5.9, −5.1]
(3) Consumer Surplus (million $)	65,472	77,298 [78199, 76404]	−11,827 [−12727, −10932]
(4) Producer Surplus (million $)	12,937	1,707 [810, 2594]	11,230 [12126, 10343]
(5) Social Welfare = (3) + (4)	78,409	79,006 [79010, 78997]	−597 [−601, −589]

Notes: The 95% confidence intervals are in brackets (based on those of the demand estimates). Prices and surpluses are expressed in the 2007 constant U.S. dollars.

For the collusion period we assume that firms compete as Cournot to obtain counterfactuals P^{CF} and q_i^{CF} (and compute profits and other surpluses). Compute the difference during the collusion period between the observed outcomes and the competitive counterfactuals.

[Igami, 2015]

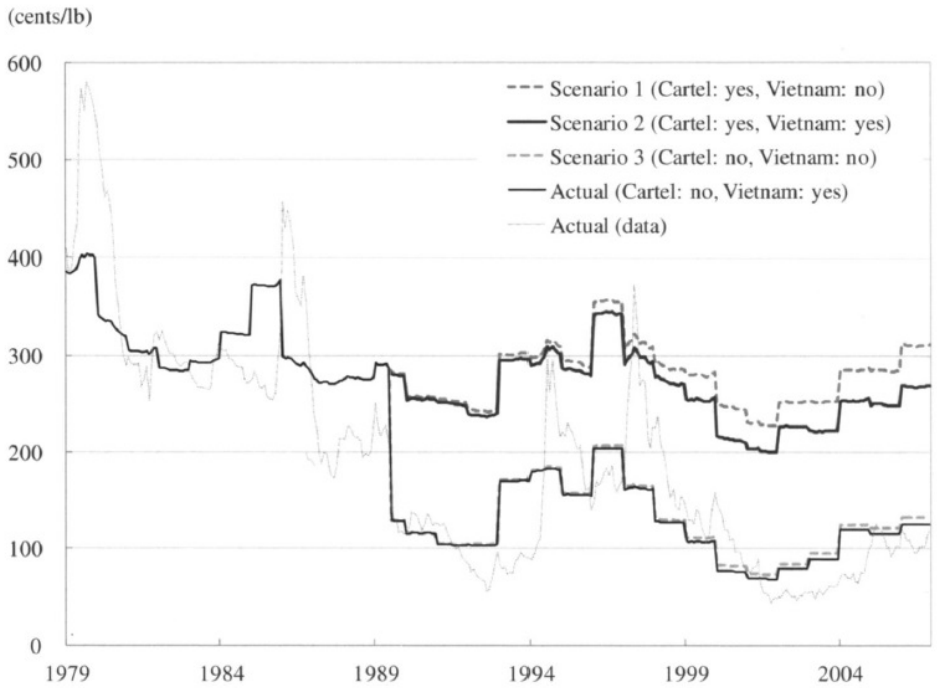


Figure 3
Counterfactual Prices

Notes: *Scenario 1* is the most profitable case for the coffee exporters based on (1) the traditional exporters' maintaining the market power under ICA even after July, 1989, and (2) zero exports leaving Vietnam in all years. *Scenario 2* assumes only (1). *Scenario 3* assumes only (2). The *Actual* case witnessed both the lapse of ICA and Vietnam's export expansion.

Finally, counterfactual prices post collusion.

Exercise 4: Motivation

- We know *when* the cartel operated but not *how* — joint profit maximisation, quotas, or something in between.
- **Why not recover MC from the collusive period?** The Cournot FOC is violated during collusion — inverting it confounds MC with the collusive markup.
- **Competitive period rescues identification:** FOC holds during competition $\Rightarrow$ recover true MC_i, then carry it into the collusive period to compute the cartel overcharge.

$\Rightarrow$ **What you learn:** cartel damages without taking a stand on how profits were split — only the competitive benchmark is needed.

Exercise 4: Quantifying collusion damages (without knowing how firms collude)

- Assume we know specific periods of competition and collusion.
- Collusion can be implemented in different ways.
- After estimating demand, recover marginal costs only using competitive period.
- Make assumption about collusive marginal costs with recovered data.
- Compare observed collusive values and a CF of competition.

[Bergquist and Dinerstein, 2020]

- RCT + Structural Estimation!!!
- Market reforms (in agriculture) in developing countries did not fully result in lower and stable prices. Why?
- Potential causes: transaction costs, transport costs, consumer search, contractual risk, etc.
- But also form of competition among intermediaries.
- Main logic
  * (Under low market power) $\downarrow$ Transaction costs, $\downarrow$ Prices.
  * (Under high market power) $\downarrow$ Transaction costs, efficiencies captured by intermediaries.

[Bergquist and Dinerstein, 2020]

- This paper: 3 RCTs in an African agricultural market.
  1. Exogenous reduction in traders' marginal costs by offering all traders in a market a subsidy.
  2. Offer consumers random reductions in price.
  3. Exogenous incentives to make traders' enter markets (randomly).
- Use data to model Cournot model with conduct parameter.

[Bergquist and Dinerstein, 2020]

- Model of supply and demand for a homogeneous product.
- Experiment 1 (identifies passthrough) and Experiment 2 (identifies demand curvature) allow to infer the model of competition under which traders operate.
- Experiment 3 check the effect of the number of traders in how these compete.

[Bergquist and Dinerstein, 2020]

Panel A. Experiments

Experiment	Exogenous variation		Used to identify
Cost shock	• Marginal costs* • Price • Marginal costs in multimarket traders' other markets*	→ → →	• Pass-through • Demand (quantity response) • MC slope
Demand	• Price*	→	• Demand (quantity response)
Entry	• Size of fixed payment to enter* • Number new traders in market	→ →	• Fixed costs of entry • Effect of entry on competition

* = directly manipulated

[Bergquist and Dinerstein, 2020]

Panel B. Simple model

Identified by cost shock experiment

Identified by demand experiment

Profit weight

$$\rho_{mw} \equiv \frac{\partial P_{mw}}{\partial c_{mw}} = \left\{ 1 + \frac{1 + E_{mw}}{N_{mw}} \left(1 + \omega \left(N_{mw} - 1 \right) \right) \right\}^{-1}$$

Observed in data

[Bergquist and Dinerstein, 2020]

Panel C. General model

Observed in data

$$P_{mw} - \Delta c_{mw} + \frac{\partial P_{mw}}{\partial q_{jmw}} q_{jmw} = -\omega \frac{\partial P_{mw}}{\partial q_{jmw}} \sum_{k \neq j} q_{kmw} + \gamma q_{jw} + c_j + c_m + c_w + c_{jmw}$$

Manipulated by cost shock experiment

Calculated from demand model (identified by cost shock and demand experiments)

Identified by instruments:

- Cost shock low
- Cost shock high
- Mean treatment status in other markets
- Entry offer

[Bergquist and Dinerstein, 2020]

- Household i, in week w and in local market m, demands $q_{imw}(P_{mw}) \geq 0$ kilograms of maize at market price P_{mw}. Given the set I_{mw} households in local market m in week w, market demand is $Q_{mw}(P_{mw}) = \sum_{i \in I_{mw}} q_{imw}(P_{mw})$ with inverse $P_{mw}(Q_{mw})$.

- Then, trader j's maximization

$$\max_{(q_{jmw})_{m \in M_{jw}}} \pi_{jmw} = \sum_{m \in M_{jw}} P_{mw}(q_{jmw}) q_{jmw} - C_{jw}(q_{j1w}, \ldots, q_{jMw})$$

$$+ w \left(\sum_{m \in M_{jw}} \sum_{k \neq j} P_{mw}(q_{jmw}) q_{kmw} - \sum_{k \neq j} C_{kw}(q_{j1w}, \ldots, q_{jMw}) \right)$$

  where M_{jw} is the set of markets j visits in week w and C_{jw} is j's total costs in week w across all markets.

- The second brackets captures rivals' profits with an internalization parameter w

[Bergquist and Dinerstein, 2020]

- Taking FOC

$$P_{mw} = \frac{\partial C_{jw}}{\partial q_{jmw}} - \frac{\partial P_{mw}}{\partial q_{jmw}} \left(q_{jmw} + w \sum_{k \neq j} q_{kmw} \right)$$

- Under symmetry and constant marginal costs

$$P_{mw} = c_{mw} - (1 + w(N_{mw} - 1)) \frac{\partial P_{mw}}{\partial Q_{mw}} \left(\frac{Q_{mw}}{N_{mw}} \right)$$

- Further assumptions:
  * All markets have the same w.
  * Interventions do not change w.

[Bergquist and Dinerstein, 2020]

- Using first and second experiment
  - Reduce c_{mw} and evaluate through passthrough

$$\rho_{mw} = \frac{\partial P_{mw}}{\partial c_{mw}} = \{1 + \frac{1 + E_{mw}}{N_{mw}} (1 + w(N_{mw} - 1))\}^{-1}$$
$$E_{mw} = (Q_{mw}/(\partial P_{mw}/\partial Q_{mw})) (\partial(\partial P_{mw}/\partial Q_{mw})/\partial Q_{mw})$$

  - Under specific models of competition

$$\rho_{mw} = \begin{cases} \{1 + \frac{1+E_{mw}}{N_{mw}}\}^{-1} & \text{, when Cournot} \\ \{2 + E_{mw}\}^{-1} & \text{, when collusive} \end{cases}$$

  - Therefore ρ_{mw} depends on w, curvature of demand E_{mw} and number of traders N_{mw}.

- Third experiment is left to the reader.

[Bergquist and Dinerstein, 2020]

- Notice what we are doing here. To get the conduct parameter you only need a passthrough and demand function estimation.

- We need very flexible demand function estimation.
  * Functional form: $P_{mt} = a - k_{mt}Q_{mt}^{\delta}$ with $E_{mt} = \delta - 1$.
  * Passthrough: $\rho_{mw} = \frac{\partial P_{mw}}{\partial c_{mw}} = \{1 + \frac{\delta}{N_{mw}}(1 + w(N_{mw} - 1))\}^{-1}$
  * Estimation exploits experiment design (using IV-GMM):

$$log(q_{im1}) - log(q_{im0}) = \frac{1}{\delta}(log(a - P_{im1}) - log(a - P_{im0})) + (log(\eta_{im1}) - log(\eta_{im0}))$$

[Bergquist and Dinerstein, 2020]

TABLE 1—PASS-THROUGH

	Price (1)	Price (2)
Cost change	0.224	
	(0.0434)	
Cost change–low		0.219
		(0.0538)
Cost change–high		0.228
		(0.0618)
Mean dependent variable	28.92	28.92
Observations	1,860	1,860
Market fixed effects	Yes	Yes
Week fixed effects	Yes	Yes

Notes: Price regressed on *Cost change*, the level of cost reduction per kg offered in market m on week w. *Cost change* is the negative value of the marginal cost subsidy in cost shock treatment markets and zero elsewhere, such that *Cost change* $=$ {0 Ksh, −200 Ksh, −400 Ksh}) or {0 US$, −1.98 US$, −3.96 US$}. Week and market fixed effects are included to improve precision. The sample includes traders in market-days in which the market was in either the cost shock treatment or control period: market days assigned to the entry treatment are omitted. Under this specification, the coefficient on the cost reduction term yields the pass-through rate, or $\partial P/\partial c$. The first column shows the overall pass-through rate of 22 percent. The second

1 cost saving unit, only reduces prices in 0.20 units.

[Bergquist and Dinerstein, 2020]

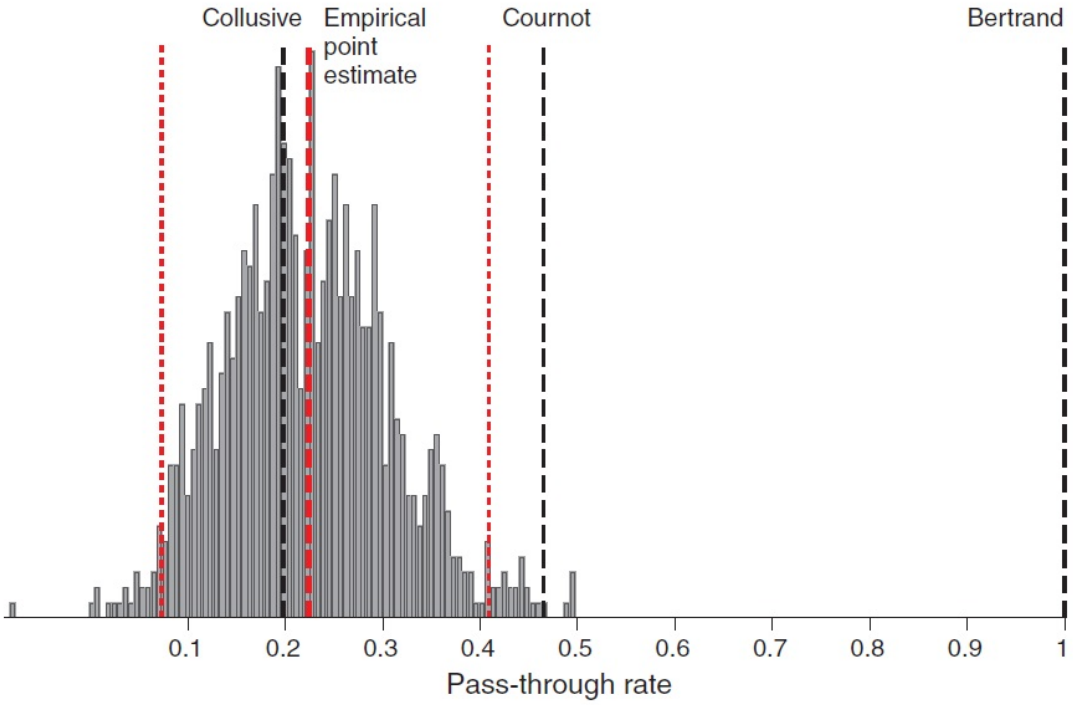
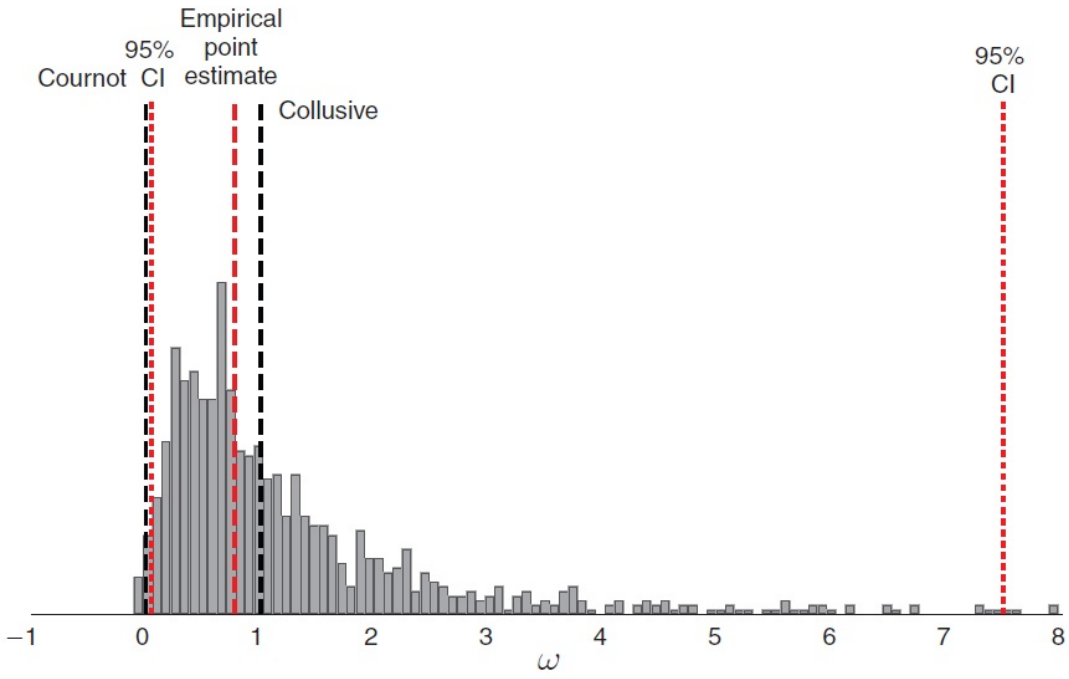


FIGURE 6. PREDICTED PASS-THROUGH UNDER THREE SIMPLE MODELS

Observed passthrough close to implied by collusion $w = 1$.

[Bergquist and Dinerstein, 2020]



w estimated close to 1.

[Bergquist and Dinerstein, 2020]

TABLE 2—MODEL ESTIMATES

	Parameter estimate	95% confidence interval lower bound	95% confidence interval upper bound
Simple model			
a	42.76	41.56	43.96
δ	4.07	1.71	6.43
ω	0.78	0.05	7.48
General model			
μ_a	29.15	28.84	52.19
σ_a	2.87	2.48	60.00
δ	4.21	0.70	9.64
ω	1.07	0.20	3.09
γ	0.0006	−0.0006	0.0016
General model, $\omega = 1$			
γ	0.0007	−0.0010	0.0016

Notes: The point estimates and the 95 percent confidence intervals for the estimated parameters of the simple and general models are displayed. The final row is the point estimate for the marginal cost slope under the general model where we impose $\omega = 1$. This is the relevant cost parameter for markup and welfare analysis, which we implement given that traders maximize joint profits. The 95 percent confidence interval on γ in the model with $\omega = 1$ imposed does not account for the testing step. For a specification that allows for joint estimation of γ and testing of the model of competition, see online Appendix Section B.

Exercise 5: Motivation

- Previous exercises estimated conduct from a supply residual — but FOC inversion is exactly where the Corts bias enters.

- **Passthrough offers an alternative:** how much of a cost shock is passed on to consumers depends on conduct.
  * Perfect competition: $\rho = 1$.
  * Cournot: $\rho < 1$, depends on N and demand curvature.
  * Collusion: ρ even lower.

- **Why cleaner?** ρ is estimated directly from cost variation — conduct enters only when you solve the passthrough expression for λ.

$\Rightarrow$ **What you learn:** conduct can be inferred without a supply residual, sidestepping the Corts bias entirely.

Exercise 5: Using passthrough to infer conduct

- Using market-level data.
- Estimate passthrough
- Estimate flexible demand
- Obtain an expression for passthrough depending on: number of firms, demand curvature and conduct.
- Solve for conduct.
- Use bootstrap to obtain distribution.

[Igami and Sugaya, 2022]

- Study of an international cartel in the vitamins market between several firms.
- Vitamins producers from different countries collude. But apparently, the cartel incentivized entry of Chinese producers (and the cartel stopped).
- Can we estimate the incentives to collude? Do they react to the entry of Chinese producers?
- Notice that the exact dates of collusion/competition are known.
  * In certain years, firms compete a la Cournot ($\lambda_{ik} = 0$).
  * Differently that [Igami, 2015], collusive periods will be explicitly modelled.

[Igami and Sugaya, 2022]

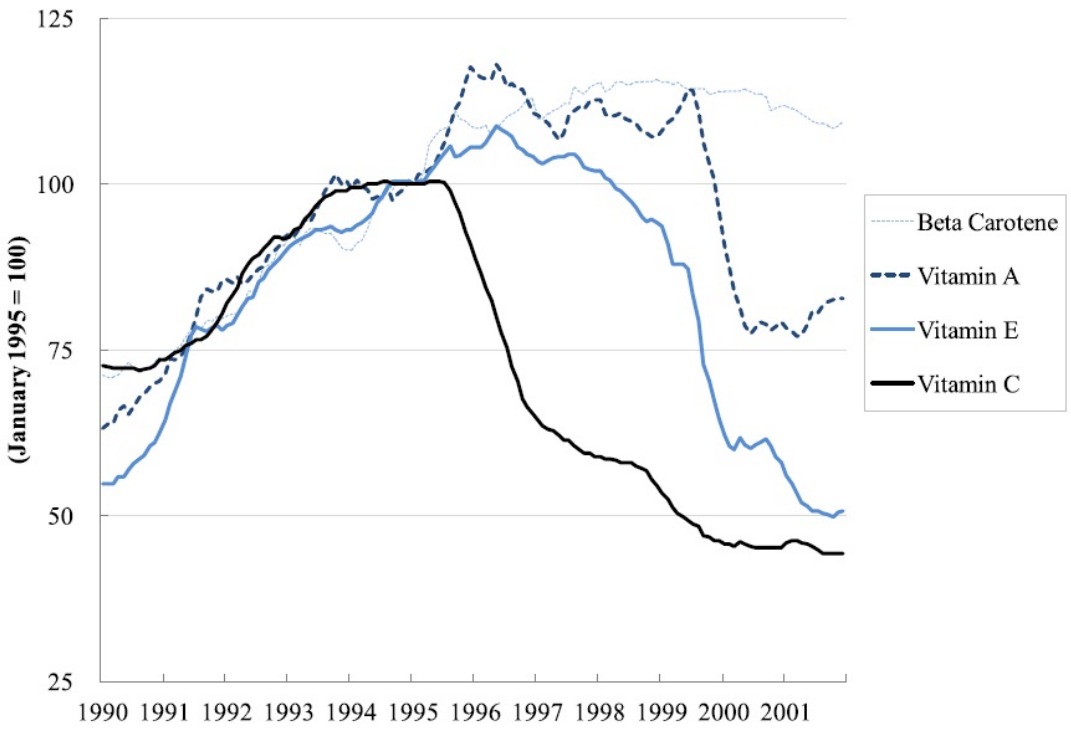


FIGURE 1

Cartels and vitamin prices

Notes: Roche's monthly average U.S. transaction prices ($/kg) are rescaled with January 1995 as the base period. The vitamin C cartel collapsed internally in August 1995, whereas the other three cartels continued operations until 1999, when the U.S. government prosecuted them.

[Igami and Sugaya, 2022]

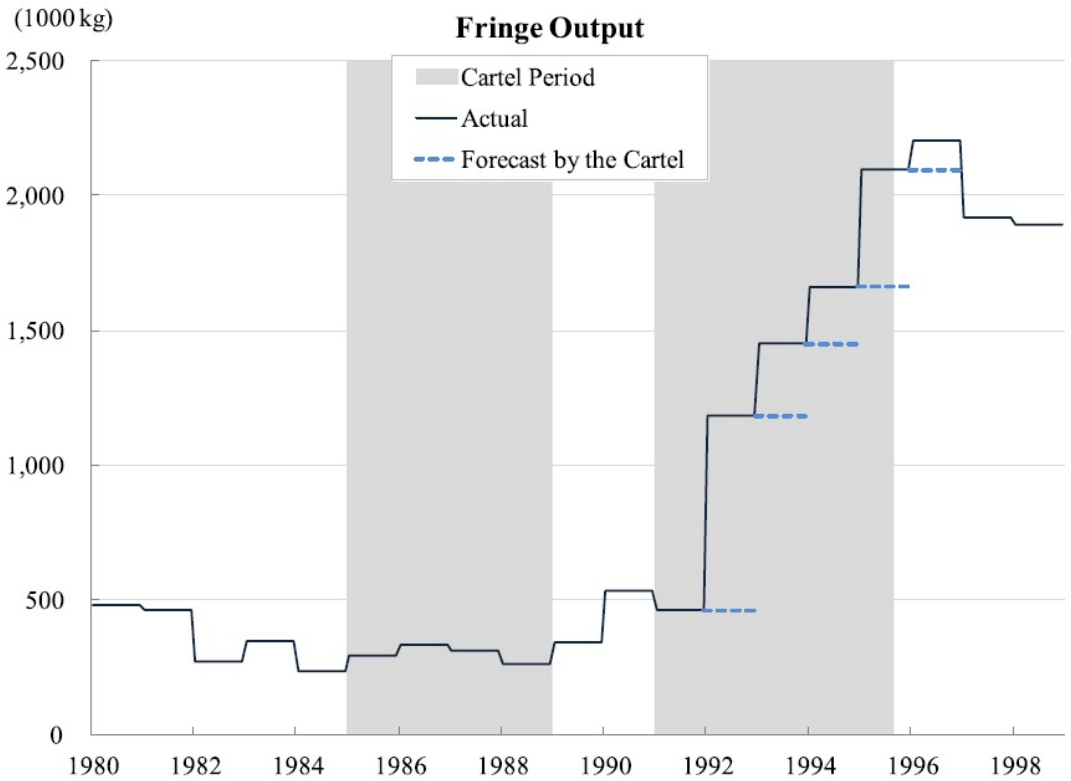


FIGURE 2

Sudden growth of fringe supply (vitamin C)

Notes: The dashed lines represent the cartel firms' static predictions, as suggested by EC (2003).
Source: Bernheim (2002a) and EC (2003).

[Igami and Sugaya, 2022]

- World Demand $Q_t = \alpha_0 + \alpha_1 P_t + \alpha_2 X_t + \epsilon_t$.
- There are fringe competitors given by $Q_{fri,t}$ and cartel quantities by $Q_{car,t}$ such that $Q_t^D = Q_{car,t} + Q_{fri,t}$.
- $Q_{fri,t}$ is specified through different specifications: $Q_{fri,t} = \kappa$, $Q_{fri,t} = \lambda_t P_t$ or $Q_{fri,t} = \kappa + \lambda_t P_t$.
- When firms are not colluding, n colluding firms behave as Cournot so:

$$P_t + \frac{dP}{dQ_t} \times q_{i,t} = c_{i,t}$$

$$c_{i,t} = c_{roche,t}^{obs} + \gamma_i + \eta_{i,t}$$

which implies

$$P_t + \frac{dP}{dQ_t} \times q_{i,t} = c_{roche,t}^{obs} + \gamma_i + \eta_{i,t}$$

for each firm i.

[Igami and Sugaya, 2022]

- Demand and supply will be estimate simultaneously with GMM for params $\theta = (\alpha_0, \alpha_1, \alpha_2, (\gamma_i)_{i=1}^n)$.

- First set of moment conditions (at yearly level)

$$\bar{m}_1(\theta) = \sum_y \bar{\epsilon}_y \cdot Z_y$$

with $\bar{\epsilon}_y = Q_y - \alpha_0 - \alpha_1 \bar{P}_y - \alpha_2 \bar{X}_y$ and $Z_y = (1, \bar{X}_y, c^{obs}_{roche,y})$. $c^{obs}_{roche,y}$ as excluded instrument for demand function.

- Second set of moment conditions (at firm, country level)

$$\bar{m}_{2,i}(\theta) = \sum_y \bar{\eta}_{i,y} \cdot W_{i,y}$$

with $\bar{\eta}_{i,y} = \bar{P}_y + (\frac{P}{Q_t}) q_{i,y} - c^{obs}_{roche,y} - \gamma_i$ and $W_{i,y} = (1, \bar{X}_y)$. $\bar{X}_y$ as excluded instrument for marginal costs. Later $\bar{m}_2(\theta) = \sum_i \bar{m}_{2,i}(\theta)$.

[Igami and Sugaya, 2022]

- Third set of moment conditions

$$\bar{m}_3(\theta) = \sum_t \bar{\eta}_t \cdot X_t$$

with $\bar{\eta}_t =$
$P_t + (1/n)\frac{dP}{dQ_t}\left(\alpha_0 + \alpha_1 P_t + \alpha_2 X_t - Q_{fri,t}\right) - c^{obs}_{roche,t} - (1/n)\sum_i \gamma_i$.

- Then, $\bar{m}(\theta) = (\bar{m}_1(\theta), \bar{m}_2(\theta), \bar{m}_3(\theta))$ and

$$\hat{\theta}_{gmm} = \text{argmin}_\theta \;\; \bar{m}(\theta)' I \bar{m}(\theta)$$

[Igami and Sugaya, 2022]

TABLE 2
GMM estimates of demand and costs (vitamin C)

Model	(1)	(2)	(3)
Specification of fringe	Time-varying intercept (Baseline) $Q_{fri,t} = \kappa_t$	Time-varying slope $Q_{fri,t} = \lambda_t P_t$	Time-invariant intercept and time-varying slope $Q_{fri,t} = Q_{fri,1990} + \lambda_t P_t$
α_0	−0.434	0.066	−0.068
	(3.375)	(2.313)	(2.330)
α_1	−0.330	−0.248	−0.312
	(0.002)	(0.002)	(0.005)
α_2	0.391	0.323	0.364
	(0.011)	(0.007)	(0.009)
γ_{roche}	0.011	−0.188	0.107
	(2.597)	(4.400)	(11.813)
γ_{takeda}	3.194	3.136	3.262
	(0.626)	(0.925)	(2.263)
$\gamma_{e.merck}$	4.474	4.426	4.485
	(0.168)	(0.234)	(0.580)
γ_{basf}	4.882	4.880	4.914
	(0.152)	(0.157)	(0.300)
Moment conditions	$\bar{m}_1, \bar{m}_2, \bar{m}_3$	$\bar{m}_1, \bar{m}_2$	$\bar{m}_1, \bar{m}_2$
Number of months	112	112	112

Notes: Standard errors in parentheses are based on 1,000 block-bootstrap samples, where each block consists of 12 consecutive months of a calendar year. See Supplementary Appendix C.3 for vitamins A and E, and beta carotene.

Each column (model) corresponds to different assumptions about fringe quantities $Q_{fri,t}$ for vitamin C.

[Igami and Sugaya, 2022]

- Stage-game profits: $\pi_{i,t} = (P_t - c_{i,t})q_{i,t}$.
- $Q_{car,t}$ is the quantity that maximizes total profit,

$$\sum_{i \in I} \left(\frac{dP}{dQ_t} \times \left(Q_{car,t} + Q_{fri,t} - \tilde{X}_t\right) - c_{i^*,t} \right) \times q_{i,t}$$

  with I as the set of cartel members, $\tilde{X}_t = \alpha_0 + \alpha_2 X_t + \epsilon_t$ and $c_{i^*,t}$ corresponds to the lowest marginal cost.
- Define a firm share as $s_{i,t}$. Then, cartel profits for firm i is $\pi^C_{i,t} = (P_t - c_{i,t})(s_{i,t}Q_{car,t})$.
- Deviation profits (max when others stick to shares) as $\pi^D_{i,t}$.
- Nash profits as $\pi^N_{i,t}$.

[Igami and Sugaya, 2022]

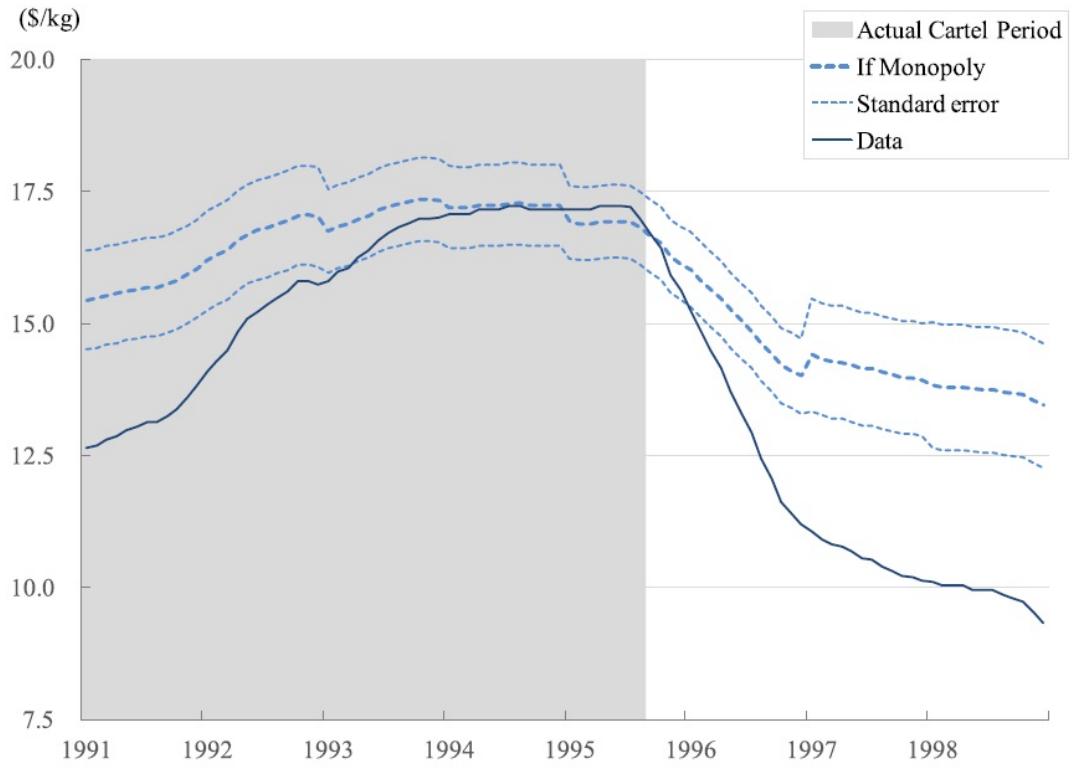


FIGURE 5

Actual and monopoly prices (vitamin C)

Notes: The hypothetical monopoly price is based on the estimates of demand and Roche's cost in column 1 of Table 2. The thin dashed lines indicate standard errors based on 1,000 block-bootstrap samples.

Observed prices and implied monopoly prices under model 1 for vitamin C.

[Igami and Sugaya, 2022]

- In each month t, each cartel firm $i \in I$ observes residual demand $(\frac{dP}{dQ_t}, \tilde{X}_t)$ and cost profile $(c_{j,t})_{j \in I}$ and decide $q_{i,t}$ conditional in private history h_i^t.

- Total supply $Q_{car,t} = \sum_{i \in I} q_{i,t}$, fringe supply $Q_{fri,t}$ and the demand (all) determine the market price $P_t = \frac{dP}{dQ_t} \times \left(Q_{car,t} + Q_{fri,t} - \tilde{X}_t\right)$.

- Quantity profile is observed with an $L = 3$ period lag: $\left((q_{j,\tau})_{j \in I}, Q_{fri,\tau}\right)_{\tau \leq t-L}$.

- Firm's i private history at beginning of month t:

$$h_i^t = \left((q_{j,\tau}, Q_{fri,\tau})_{j \in I, \tau \leq t-3}, q_{i,t-2}, q_{i,t-1}, (P_\tau)_{\tau \leq t-1}\right)$$

- And public history: $h^t = \left((q_{j,\tau}, Q_{fri,\tau})_{j \in I, \tau \leq t-3}, (P_\tau)_{\tau \leq t-1}\right)$.

[Igami and Sugaya, 2022]

- Firms have static expectations (they expect $Q_{fri,\tau} = Q_{fri,t-3}$ for each $\tau \geq t$).

- Evolution of demand and costs $\left(\left(\tilde{X}_t, \alpha_1, (c_{j,\tau})_{j\in I}\right)_{\tau=1}^{\infty}\right)$ is common knowledge and firms have rational expectations with perfect foresight (and are not part of the history).

- Cartel firms supply quota allocations $(\bar{q}_{i,\tau|t})_{i\in I,\tau\geq t}$, given rational expectations of demand and costs $\{\tilde{X}_t, \alpha_1, (c_{j,\tau})_{j\in I}\}_{\tau\geq t}$ and the static expectation of the fringe supply function for each $\tau \geq t$.

- $\tau|t$ indicates expected future cartel production plan for period $\tau \geq t$ conditional on the static expectation $Q_{fri,\tau} = Q_{fri,t-3}$ formed as of current period t.

- A deviation occurs when some firms does not produce quota for the first time in period $\tau - 3$.

[Igami and Sugaya, 2022]

- Complying with the cartel agreement from month τ on gives firm i

$$V^C_{i,\tau|t}(\beta) = \sum_{s \geq \tau} \beta^{s-\tau} \pi^C_{i,s|t}$$

- Deviation payoff

$$V^D_{i,\tau|t}(\beta) = \sum_{s=\tau}^{\tau+2} \pi^D_{i,s|t} + \sum_{s \geq \tau+3} \beta^{s-\tau} \pi^N_{i,s|t}$$

because no punishment is conducted until government statistics verify compliance $L = 3$ months later.

[Igami and Sugaya, 2022]

- Cartel stability requires that each firm's individual incentive to collude is positive,

$$\text{Incentive to collude}_{i,t}(\beta) = \min_{\tau \geq t} \left(V^{C}_{i,\tau|t}(\beta) - V^{D}_{i,\tau|t}(\beta) \right) \geq 0$$

  and if there exist $i \in I$ and $\tau \geq t$ for which $V^{C}_{i,\tau|t} < V^{D}_{i,\tau|t}$ it becomes common knowledge that at t that at τ some firm deviates. Then, each firm deviates in month t.

- Collusion will be an equilibrium if

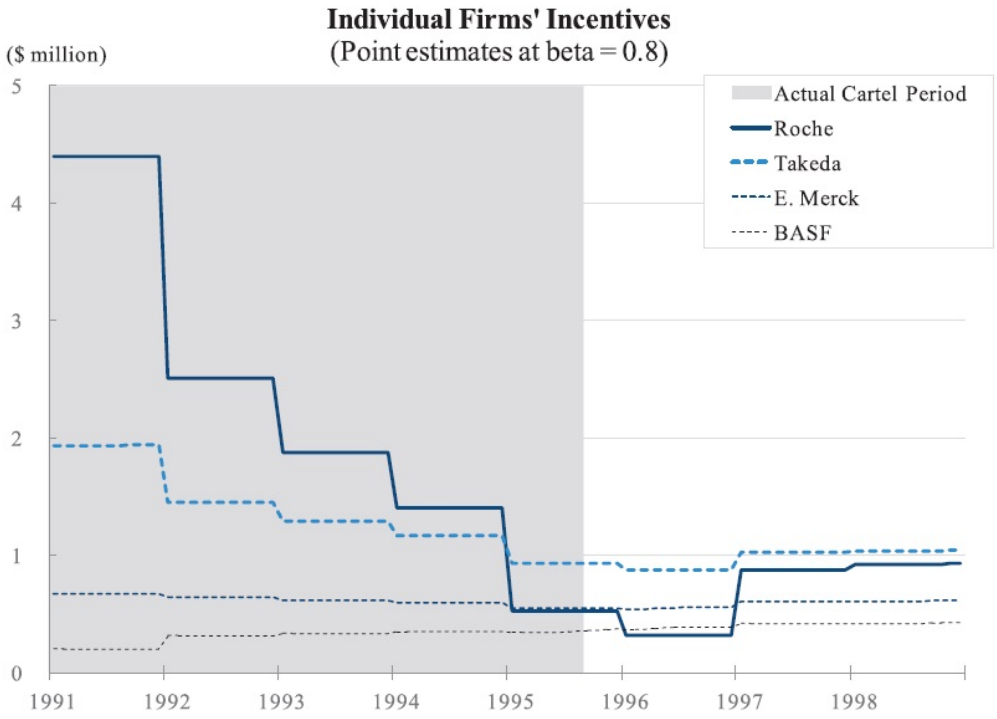
$$\text{Incentive to collude}_{t}(\beta) = \min_{i \in I} \left(\text{Incentive to collude}_{i,t}(\beta) \right) \geq 0$$

[Igami and Sugaya, 2022]

- Although the model assumes perfect monitoring (with delay), uncertainty (to the econometrician) can be incorporated through bootstrap.

- Block-bootstrap to compute θ^{bs} with $bs = 1, 2, \ldots, BS$. Under $BS = 1000$ (from demand and supply estimation) to calculate 1000 sequences of Incentive to collude$_t(\theta^{bs}; \beta)$

[Igami and Sugaya, 2022]



Largest firm (Roche) decreased coinciding with cartel breakup. Cost asymmetry increases effect of larger $Q_{fri,t}$.

[Igami and Sugaya, 2022]

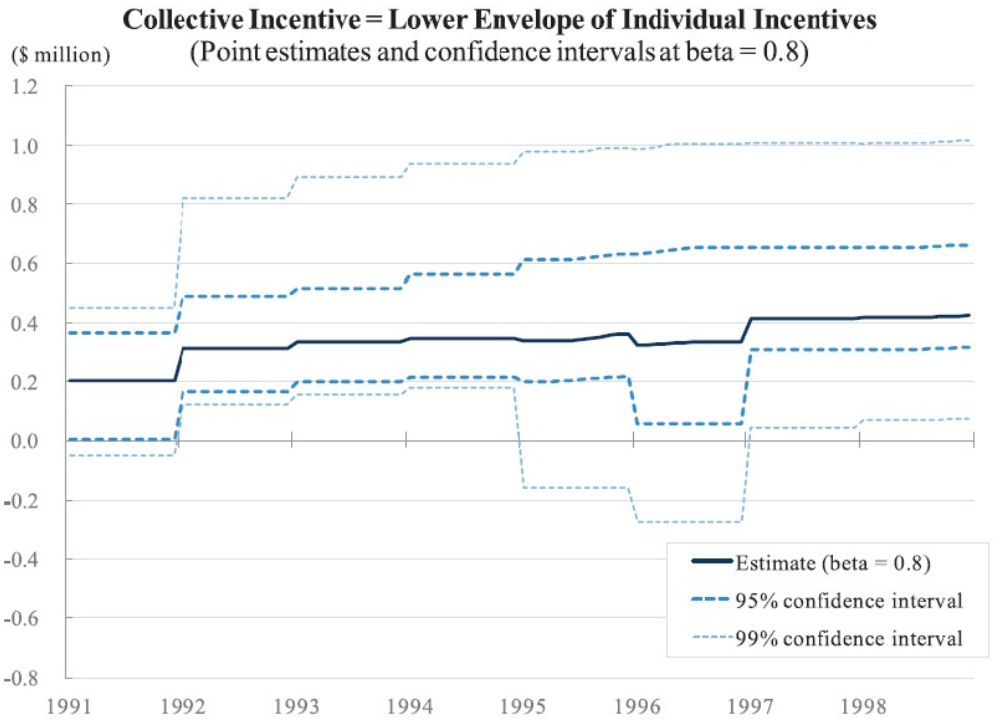


FIGURE 6

Individual and collective incentives to collude (vitamin C)

Notes: See equations (13) and (14) for the definitions of the "incentives to collude." All values are rescaled as the average monthly profits (*i.e.* multiplied by $1-\beta$). We assume the demand, the costs, and fringe supply after the sample period remain constant at their end-of-sample values in December 1998. The confidence intervals in the bottom panel reflect the 99.5th, 97.5th, 2.5th, and 0.5th percentiles of the 1,000 block-bootstrap estimates.

Same but over all firms. Notice that confidence interval includes 0.

[Igami and Sugaya, 2022]

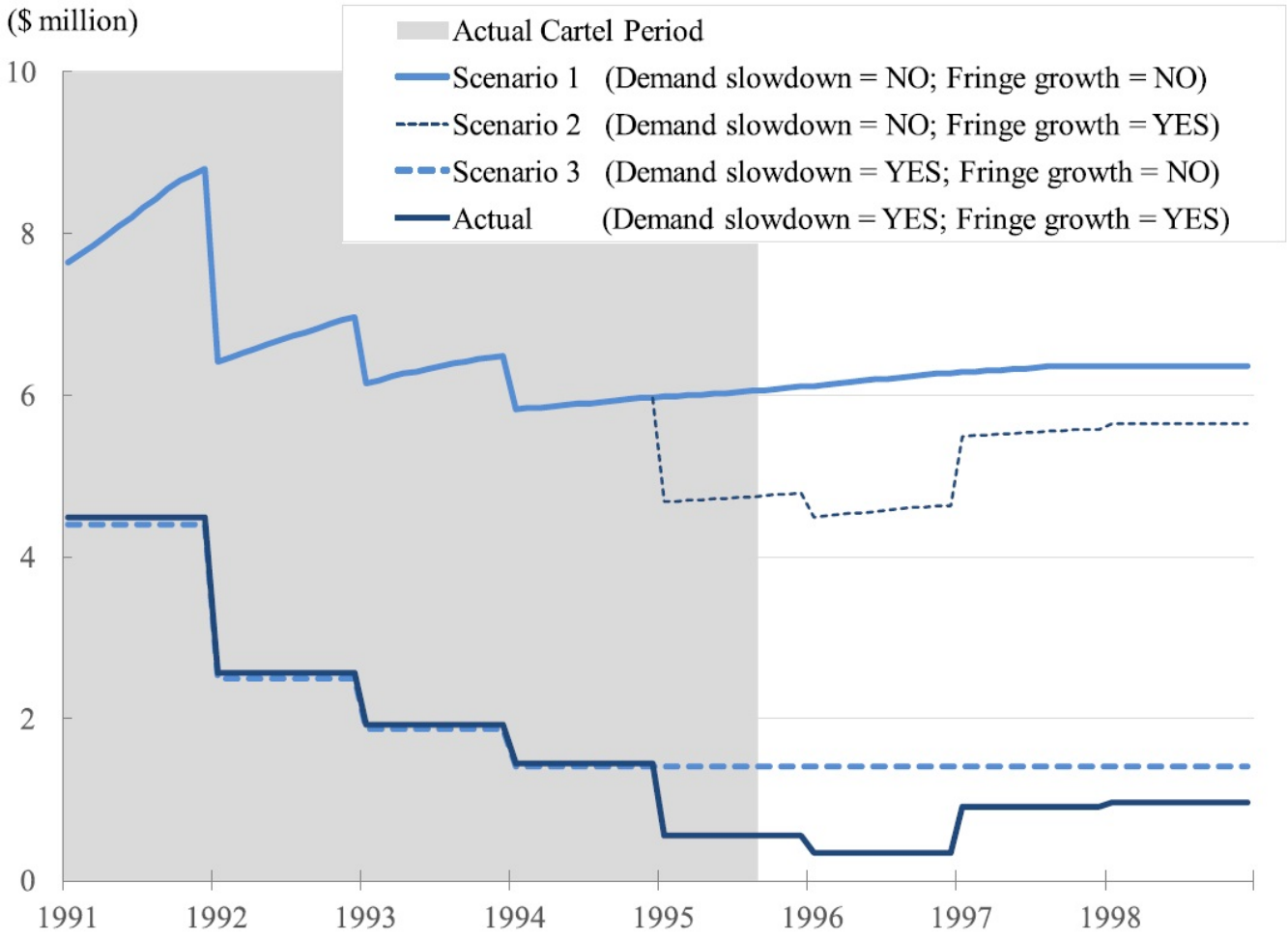


FIGURE 8

Effects of demand and supply on Roche's incentive (vitamin C)

Impact of market slowdown is larger than the increase in $Q_{fri,t}$ for cartel stability.

[Igami and Sugaya, 2022]

- How a merger affects coordinated effects?
  * *Direct market-share effect* (Positive)
  * *Deviating mitigating effect* (Positive)
  * *Convergence effect* (Negative)
  * *Cost profile (efficiencies) effect* (Depends)
- Compare how incentives to collude can change under different cost efficiencies.
- Also check if a merger would be profitable for involved firms.

[Igami and Sugaya, 2022]

- Merge between BASF-Takeda.
  * Assume it was in 1990 (one year before the cartel started) instead of 2001.
  * Once merged, BASF shuts-down facilities and all is produced by Takeda.
  * All other firms' marginal costs remain unchanged.
  * Possible cost efficiencies by $\sigma \in [0, 1)$ and

$$c^{post}_{basf,\tau|t} = (1-\sigma) \times \min\{c^{pre}_{takeda,\tau|t}, c^{pre}_{basf,\tau|t}\}$$
$$(1-\sigma) \times c^{pre}_{takeda,\tau|t}$$

[Igami and Sugaya, 2022]

TABLE 3

Cartel stability under hypothetical BASF-Takeda Merger in 1990

Merger scenario	No merger	BASF-Takeda merger								
Synergy (σ)		0	0.05	0.1	0.15	0.2	0.25	0.3	0.4	0.5
Collusive incentive	329	485	512	483	450	411	367	318	202	64
Coordinated effect	0	156	183	154	121	82	38	−12	−127	−264
(% change)	±0%	+47%	+55%	+47%	+37%	+25%	+11%	−4%	−31%	−80%

Notes: The numbers (in thousand dollars) indicate the point estimates of the cartel' s collective incentive in equation (14) as of August 1995 under $\beta = 0.8$. The first column shows our baseline estimates without merger. The other columns show results under the counterfactual BASF-Takeda merger in 1990 with specific levels of efficiency gain ("synergy"). See equation (22) for the definition of synergy, σ.

Once cost efficiencies increase, coordinated effects become negative.

[Igami and Sugaya, 2022]

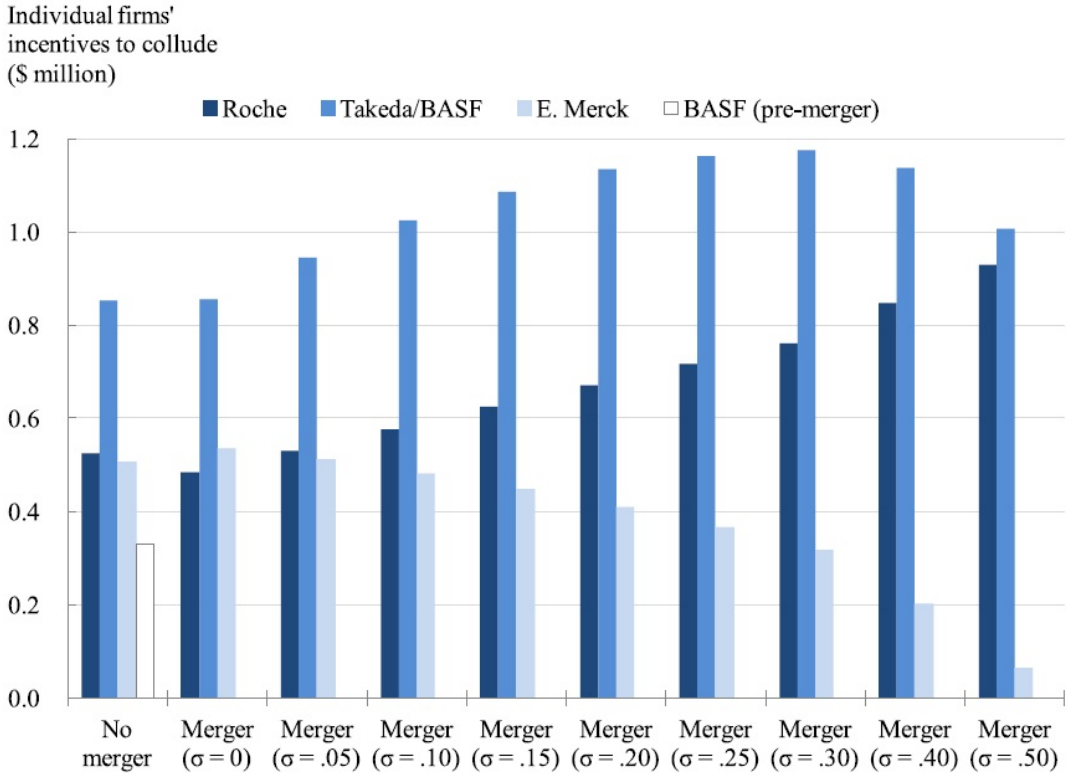


FIGURE 9
Each firm's incentive at different levels of BASF-Takeda synergy

Asymmetric effect for firms depends if they are low(Roche) - high(E.Merck) cost. Also notice effect is non-monotonic for Takeda/BASF.

Exercise 6: Motivation

- Exercises 4–5 measured *what* collusion did. Exercise 6 asks *why* firms colluded and when it breaks down.

- **IC condition:** a firm stays in the cartel as long as

$$V_i^C(\beta) - V_i^D(\beta) \;\geq\; 0$$

  Once demand and costs are estimated, this becomes a **testable restriction** on the profit structure.

- β^* **interpretation:** minimum patience for collusion to be self-sustaining. High $\beta^* \Rightarrow$ fragile cartel; low $\beta^* \Rightarrow$ robust to impatience.

$\Rightarrow$ **What you learn:** whether the profit structure supports collusion and which firm has the strongest incentive to defect.

Exercise 6: Estimation of incentives to collude

- Assume we know specific periods of competition and collusion.
- Required inputs:
  1. L and β.
  2. Define π_i^C as a fraction of π^m given the demand estimation and known or assumed costs.
  3. Define π_i^D with same information as used in π_i^C. Check if capacity constraints exist.
  4. Define π_i^N with same information as used in π_i^C. Use Nash reversion as a simple approach.
  5. In summary: (L, β), demand estimation and data/estimates of costs.

Exercise 6: Estimation of incentives to collude

- Deliverable outputs:
  - Incentives to collude under specific β.
  - Critical discount factor β^* (that makes IC equal to 0).
  - The highest possible attainable cartel price $\bar{P}$.

Exercise 6: Estimation of incentives to collude

- Summary of procedures (under no expectations):
  1. For each firm i, calculate the present value of cooperation $V_i^C = \left(\frac{1}{1-\beta}\right)\pi_i^C$
  2. Same for $V_i^D = \sum_{t=0}^{L-1}\beta^t\pi_i^D + \sum_{t=L}^{\infty}\pi_i^N$ (if $L > 1$) or $V_i^D = \pi_i^D + \left(\frac{\beta}{1-\beta}\right)\pi_i^N$
  3. Define

$$\begin{aligned}\text{Incentive to collude}_i(\beta) = &\quad V_i^C - V_i^D\\ \text{Incentive to collude}(\beta) = &\quad \min_{i\in I}\text{Incentive to collude}_i(\beta) \geq 0\end{aligned}$$

  4. To make results probabilistic use bootstrap.

$$\Pr(\text{collusion};\beta) = \frac{1}{BS}\sum_{bs=1}^{BS} 1 \times \{\text{Incentive to collude}(\theta^{bs}\beta) \geq 0\}$$

The Corts Critique (Firm-level) - Setup

- Inverse demand (market t): $P_t = A_t - b_t Q_t, \quad Q_t = \sum_{i=1}^{N} q_{it}$.
- Step 1 (demand) provides $\{A_t, b_t\}$ or elasticities; treat them as **known** in Step 2.
- Firm i's marginal cost: $mc_{it} = x_{it}^c \gamma + u_{it}$, with observed cost shifters x_{it}^c, unknown γ, and unobserved u_{it}.
- **Conduct (rival internalization):** firm fully internalizes own q_{it} and a fraction $\theta \in [0, 1]$ of rivals' output.

The Corts Critique (Firm-level) - Estimation

FOC (your S'):

$$P_t \;=\; mc_{it} \;+\; b_t\big(q_{it} + \theta\, Q_{-i,t}\big), \qquad Q_{-i,t} := Q_t - q_{it}.$$

Subtract observed cost shifters:

$$\underbrace{P_t - x_{it}^c \gamma}_{y_{it}} \;=\; b_t\, q_{it} \;+\; b_t\, \theta\, Q_{-i,t} \;+\; \underbrace{u_{it}}_{\text{supply residual}} \quad . \tag{1}$$

- Estimate (γ, θ) by GMM/2SLS using **cost instruments** Z_{it}^c.
- Demand parameters (A_t, b_t) are taken as given from Step 1.

The Corts Critique (Firm-level) - Why?

Even without market aggregation, a **constant** θ can misrepresent conduct because the residual absorbs state dependence:

1. **State-varying conduct** θ_t (e.g., tighter states $\Rightarrow$ more internalization):

$$y_{it} \;=\; b_t q_{it} + b_t \theta_t Q_{-i,t} + u_{it} \quad \Rightarrow \quad \kappa_{it} := b_t(\theta_t - \theta)\, Q_{-i,t}.$$

2. **Omitted/mismeasured costs** u_{it} that *co-move* with tight demand (overtime, capacity congestion, inventories) $\Rightarrow$ $\mathrm{Cov}(u_{it}, b_t Q_{-i,t}) \neq 0$.
3. **Dynamics/tacit collusion** in tight states: static FOC misses continuation values $\Rightarrow$ static θ mimics higher markups.

Result: the estimator shifts $\widehat{\theta}$ to soak up systematic, state-dependent variation left in the residual.

The Corts Critique (Firm-level) - *Levels* vs *Slopes*

Define the (model-implied) markup net of own output:

$$m_{it} \equiv P_t - mc_{it} - b_t q_{it} = b_t \theta Q_{-i,t} + \kappa_{it}, \quad \kappa_{it} := u_{it} + b_t(\theta_t - \theta) Q_{-i,t}.$$

Estimating θ by regressing m_{it} on $b_t Q_{-i,t}$ with instruments Z identifies a **slope**:

$$\widehat{\theta} = \frac{\mathrm{Cov}\big(\Pi_Z[m_{it}],\, \Pi_Z[b_t Q_{-i,t}]\big)}{\mathrm{Var}\big(\Pi_Z[b_t Q_{-i,t}]\big)} = \theta_{\text{true}} + \frac{\mathrm{Cov}\big(\Pi_Z[\kappa_{it}],\, \Pi_Z[b_t Q_{-i,t}]\big)}{\mathrm{Var}\big(\Pi_Z[b_t Q_{-i,t}]\big)}.$$

- **Theory (conduct)**: θ_{true} governs the *level* of markups in a given state.
- **Estimator**: $\widehat{\theta}$ measures the *reaction* (slope) of markups to instrument-induced fluctuations in $b_t Q_{-i,t}$.

The Corts Critique (Firm-level) - When doesn't matter?

- **Knife-edge:** constant θ, fixed b_t, correctly specified costs, and instruments orthogonal to κ_{it}.
- **Practical mitigation:**
  - Allow θ to vary with observables: $\theta_t = \rho' Z_t^{\text{state}}$ (season, capacity, post-merger/remedies).
  - Enrich costs: include the same state drivers that move elasticities/tightness; allow convexity.
  - Use **pure cost instruments** for Step 2 (do not reuse demand shifters that may shift elasticity).

Diagnostics & Reporting

- Over-ID tests for constant θ.
- Split-sample (tight vs. slack states): large swings in $\widehat{\theta} \Rightarrow$ Corts warning.
- Plot supply residuals $\widehat{\xi}^s_{it}$ vs. demand tightness (A_t, b_t) and vs. $b_t Q_{-i,t}$.
- Report both $\widehat{\theta}$ and the implied $h_{\text{eff}} = 1 + \widehat{\theta}(N-1)$ (for intuition).

Main takeaways

- Market-level data only allow us to do as much. We can get an approximate idea of competitiveness.
- Conduct estimation is usually underestimated but any additional information about costs helps.
- Once we have firm-level data, more ideas can be explored (like mergers, incentives to collude, etc). Under homogeneous goods is difficult to have variation to identify conduct.
- A key aspect is if we know the competitive/collusive periods.
- Usual robustness checks: demand functional form, linear or constant MC, weak IV tests, bootstrapping, etc.

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